



LAB 1

CONSTRUCT A SIMPLE NETWORK

Name: Nguyen Minh Nhut

ID: B2205896

Group: M02

Submission: an ID_NAME_Lab01.pdf file describes clearly how did you solve the problem

Exercise 0: change the directory to your home directory

Answer: \$cd

```
lnk@lnk: ~$ cd
lnk@lnk: ~$ |
```

Exercise 1: Construct a simple network with two hosts connected to the same collision domain.

Step 1 - creating the vms:

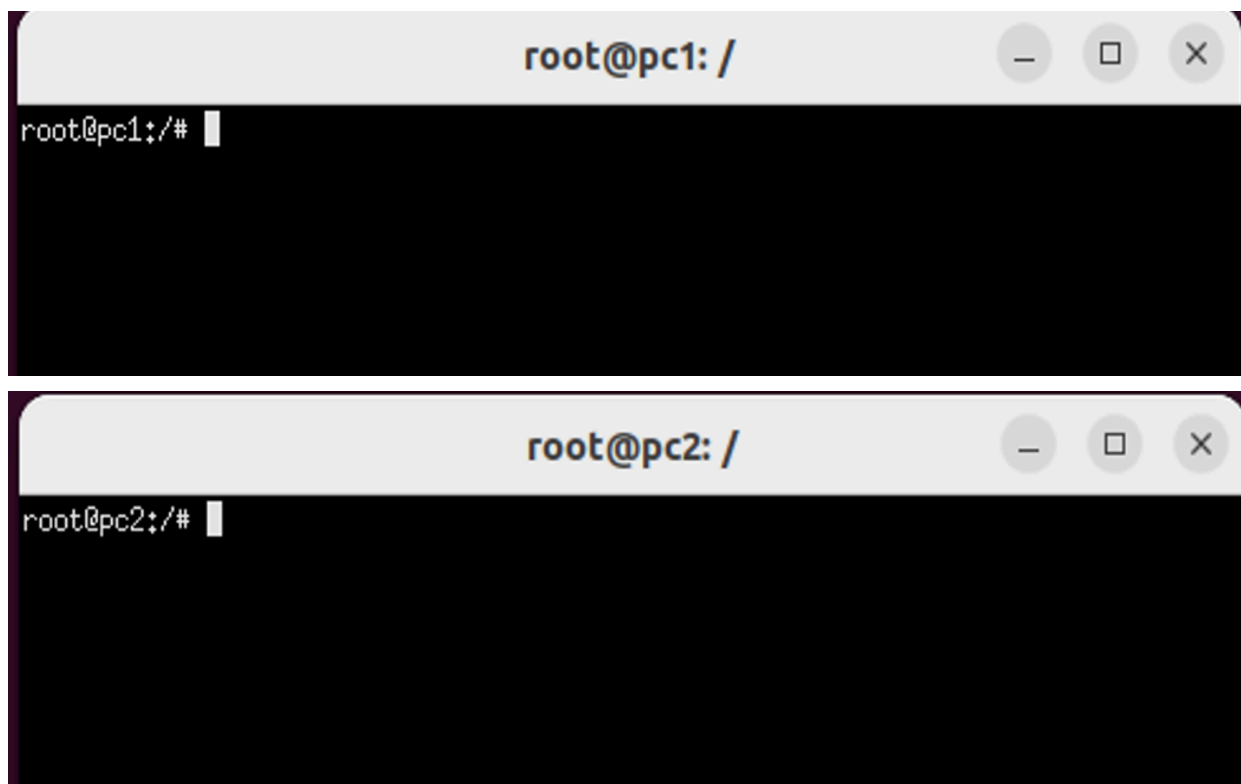
Answer:

\$ kathara vstart --eth 0:A -n pc1

\$ kathara vstart --eth 0:A -n pc2

```
nmn@Ubuntub2205896: ~$ kathara vstart --eth 0:A -n pc1
Starting Device `pc1`
[Deploying collision domains]
[Deploying devices]
nmn@Ubuntub2205896: ~$ kathara vstart --eth 0:A -n pc2
Starting Device `pc2`
[Deploying collision domains]
[Deploying devices]
nmn@Ubuntub2205896: ~$
```

Result

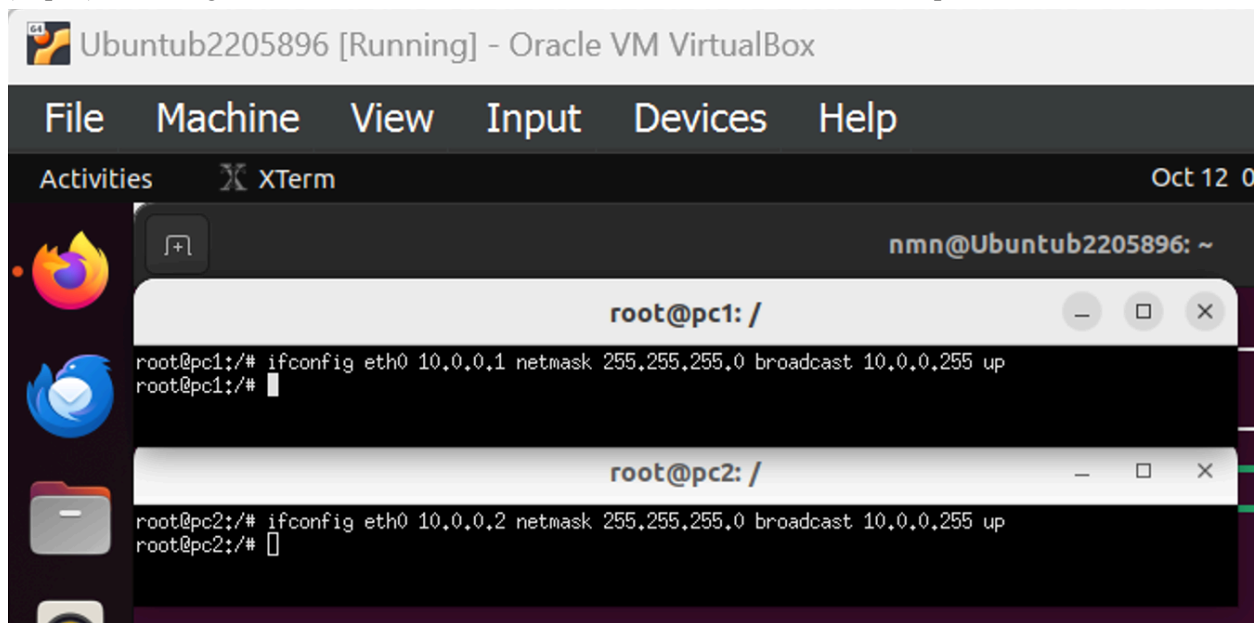


Step 2 - configuring network interfaces

Answer:

(In pc1) # ifconfig eth0 10.0.0.1 netmask 255.255.255.0 broadcast 10.0.0.255 up

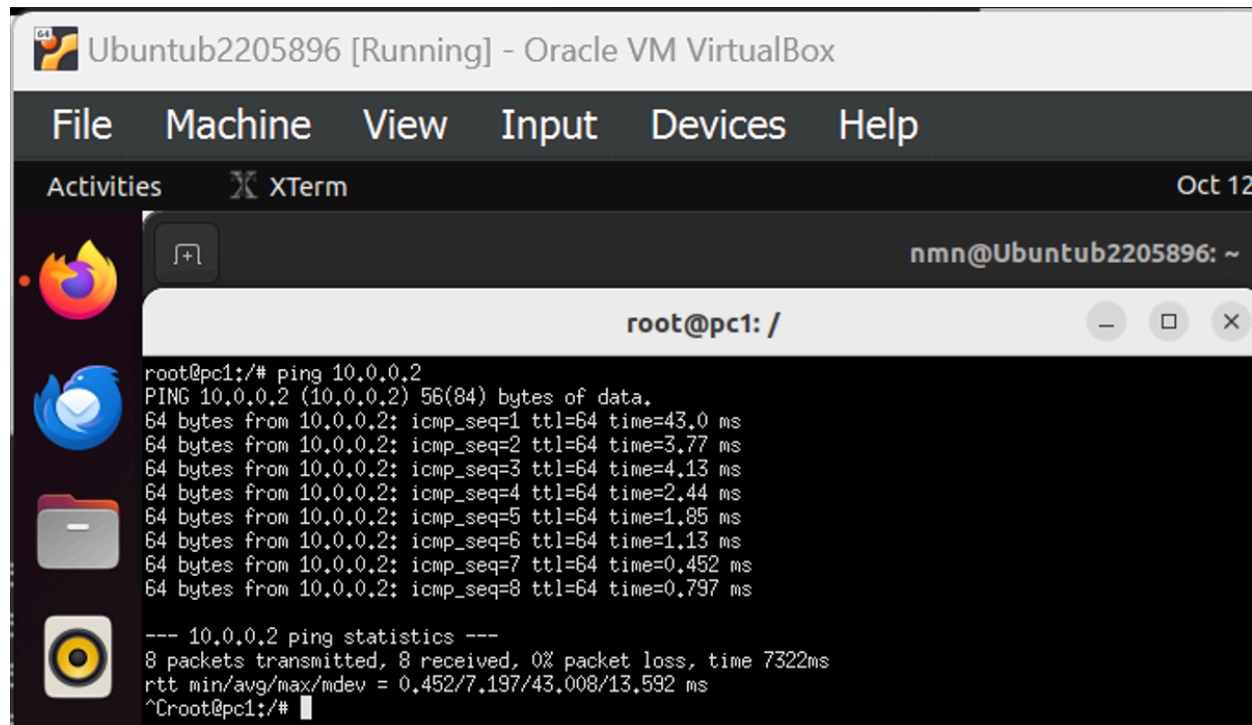
(In pc2) # ifconfig eth0 10.0.0.2 netmask 255.255.255.0 broadcast 10.0.0.255 up



Step 3 - ping

Answer:

(In pc1) # ping 10.0.0.2 (Press Ctrl + C to stop ping process)

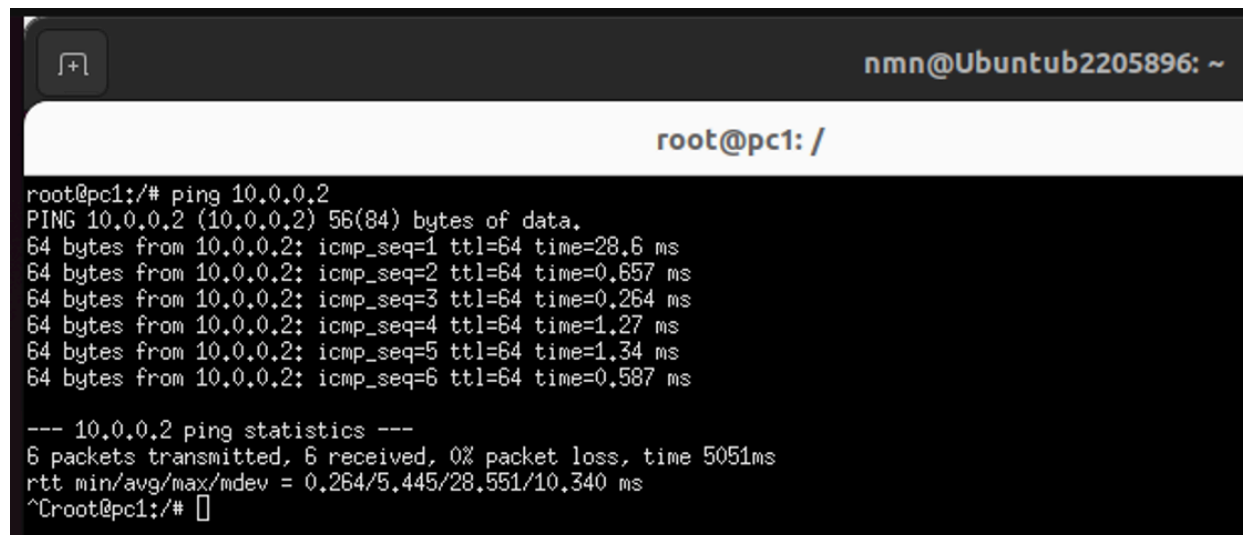


```
nmn@Ubuntub2205896: ~  
root@pc1: /  
root@pc1:/# ping 10.0.0.2  
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.  
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=43.0 ms  
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=3.77 ms  
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=4.13 ms  
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=2.44 ms  
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=1.85 ms  
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=1.13 ms  
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.452 ms  
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.797 ms  
  
--- 10.0.0.2 ping statistics ---  
8 packets transmitted, 8 received, 0% packet loss, time 7322ms  
rtt min/avg/max/mdev = 0.452/7.197/43.008/13.592 ms  
^Croot@pc1:/#
```

Step 4a - a look at the packets

Answer:

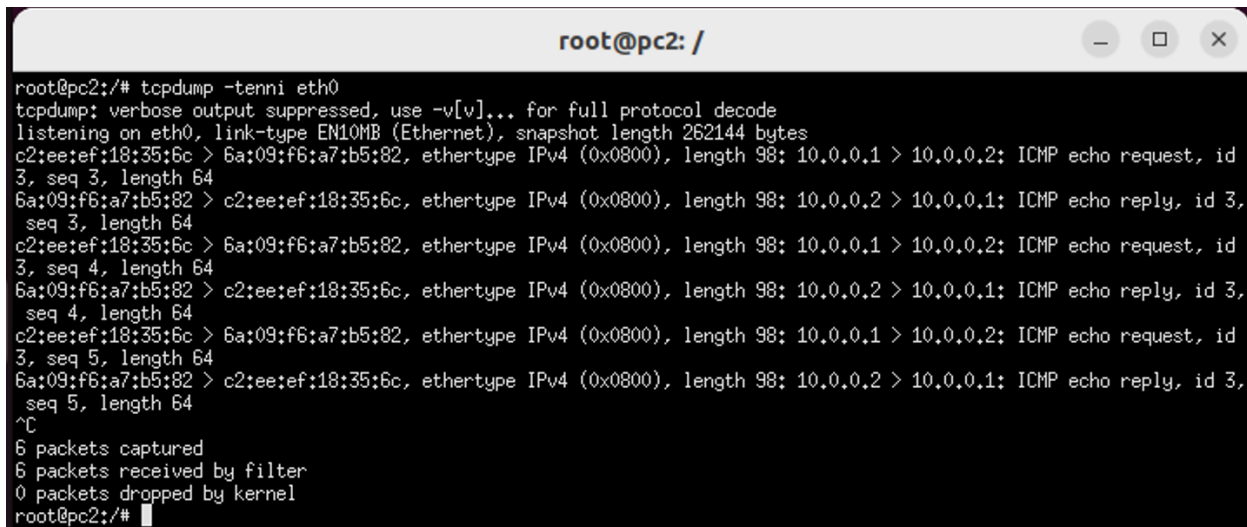
(In pc1) # ping 10.0.0.2 (Press Ctrl + C to stop ping process)



```
nmn@Ubuntub2205896: ~  
root@pc1: /  
root@pc1:/# ping 10.0.0.2  
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.  
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=28.6 ms  
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.657 ms  
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.264 ms  
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=1.27 ms  
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=1.34 ms  
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.587 ms  
  
--- 10.0.0.2 ping statistics ---  
6 packets transmitted, 6 received, 0% packet loss, time 5051ms  
rtt min/avg/max/mdev = 0.264/5.445/28.551/10.340 ms  
^Croot@pc1:/#
```

(In pc2 at the same time)

tcpdump -ttni eth0 (Ctrl + C to interrupt)



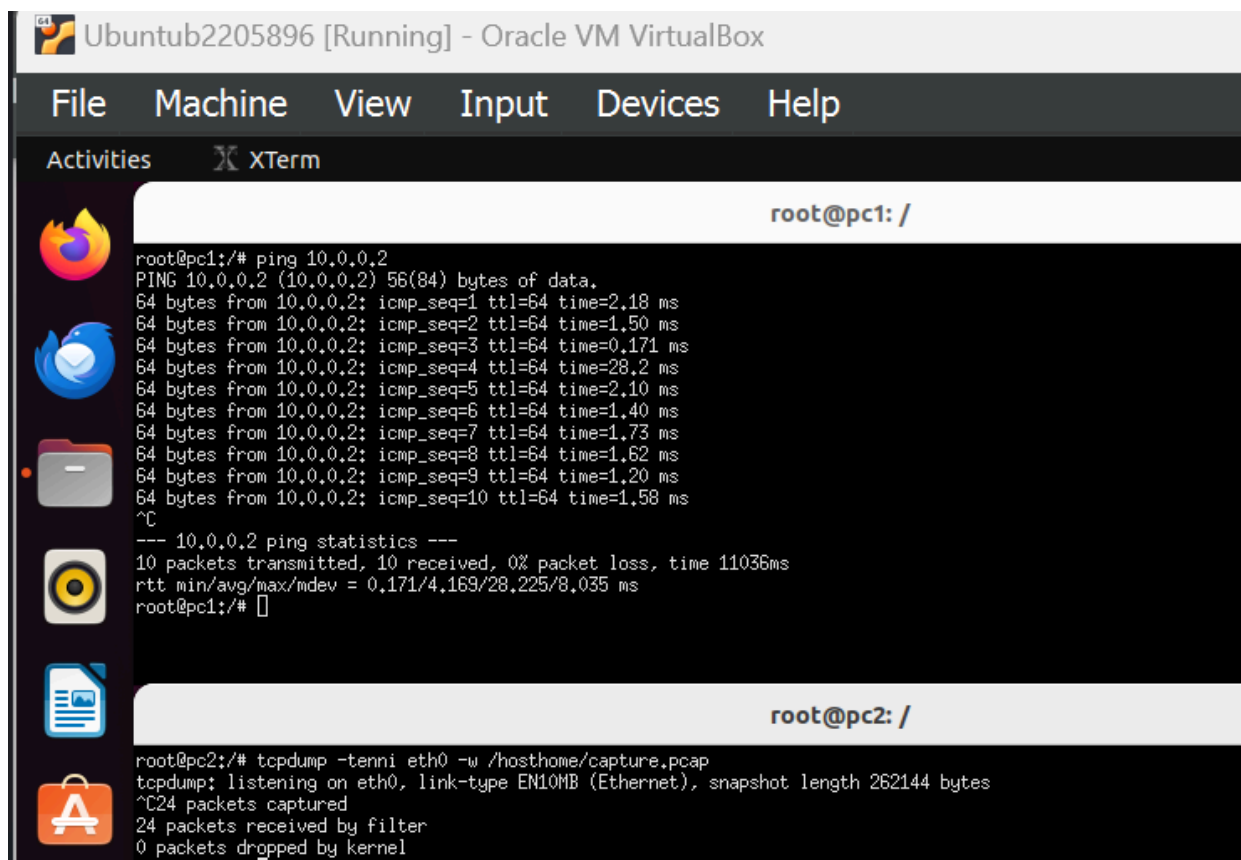
```
root@pc2: /
root@pc2:/# tcpdump -ttni eth0
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
c2:ee:ef:18:35:6c > 6a:09:f6:a7:b5:82, ethertype IPv4 (0x0800), length 98: 10.0.0.1 > 10.0.0.2: ICMP echo request, id 3, seq 3, length 64
6a:09:f6:a7:b5:82 > c2:ee:ef:18:35:6c, ethertype IPv4 (0x0800), length 98: 10.0.0.2 > 10.0.0.1: ICMP echo reply, id 3, seq 3, length 64
c2:ee:ef:18:35:6c > 6a:09:f6:a7:b5:82, ethertype IPv4 (0x0800), length 98: 10.0.0.1 > 10.0.0.2: ICMP echo request, id 3, seq 4, length 64
6a:09:f6:a7:b5:82 > c2:ee:ef:18:35:6c, ethertype IPv4 (0x0800), length 98: 10.0.0.2 > 10.0.0.1: ICMP echo reply, id 3, seq 4, length 64
c2:ee:ef:18:35:6c > 6a:09:f6:a7:b5:82, ethertype IPv4 (0x0800), length 98: 10.0.0.1 > 10.0.0.2: ICMP echo request, id 3, seq 5, length 64
6a:09:f6:a7:b5:82 > c2:ee:ef:18:35:6c, ethertype IPv4 (0x0800), length 98: 10.0.0.2 > 10.0.0.1: ICMP echo reply, id 3, seq 5, length 64
^C
6 packets captured
6 packets received by filter
0 packets dropped by kernel
root@pc2:/#
```

Step 4b - looking at the packets with a graphical interface, same as before, but store sniffed packets into file capture.pcap (on the host machine)

Answer:

(In pc1) # ping 10.0.0.2 (Press Ctrl + C to stop ping process)

(In pc2 at the same time) # tcpdump -ttni eth0 -w /home/capture.pcap (Ctrl + C to interrupt)



```

root@pc1: /

root@pc1:~# ping 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=2.18 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=1.50 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.171 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=28.2 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=2.10 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=1.40 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=1.73 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=1.62 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=1.20 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=1.58 ms
^C
--- 10.0.0.2 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 11036ms
rtt min/avg/max/mdev = 0.171/4.169/28.225/8.035 ms
root@pc1:~#

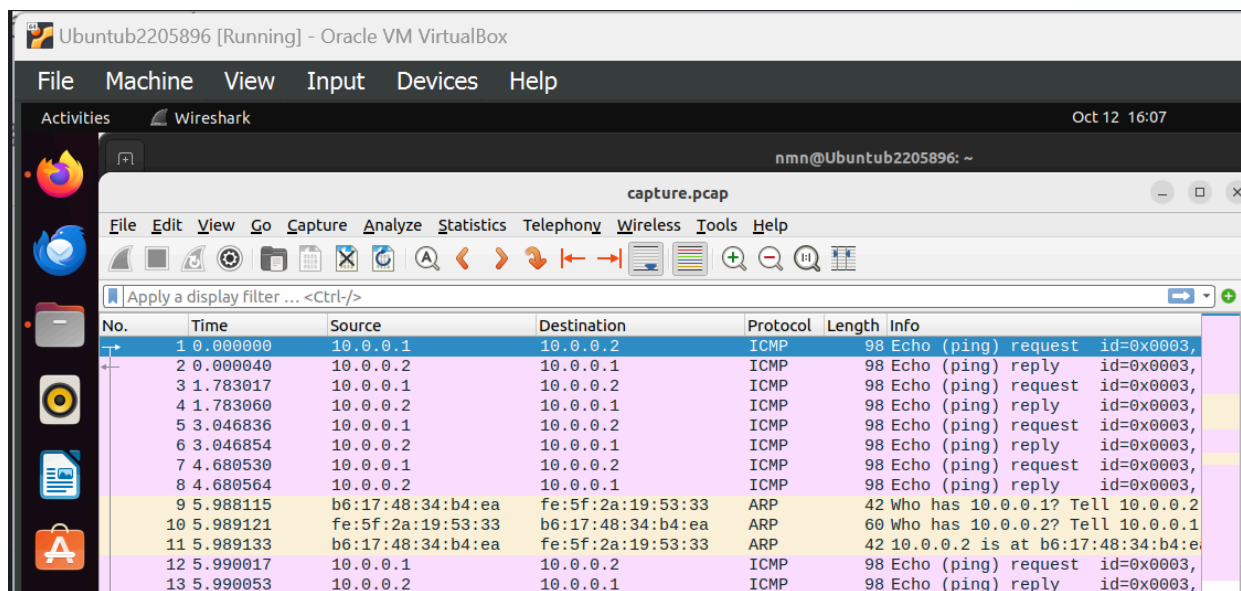
root@pc2: /

root@pc2:~# tcpdump -tnni eth0 -w /hoshome/capture.pcap
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
^C24 packets captured
24 packets received by filter
0 packets dropped by kernel
root@pc2:~#

```

Step 4 - looking at the packets with a graphical interface open capture.pcap on the real host machine using a packet dissector

Answer:



No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0x0003, seq=1
2	0.000040	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply id=0x0003, seq=1
3	1.783017	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0x0003, seq=2
4	1.783060	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply id=0x0003, seq=2
5	3.046836	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0x0003, seq=3
6	3.046854	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply id=0x0003, seq=3
7	4.680530	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0x0003, seq=4
8	4.680564	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply id=0x0003, seq=4
9	5.988115	b6:17:48:34:b4:ea	fe:5f:2a:19:53:33	ARP	42	Who has 10.0.0.1? Tell 10.0.0.2
10	5.989121	fe:5f:2a:19:53:33	b6:17:48:34:b4:ea	ARP	60	Who has 10.0.0.2? Tell 10.0.0.1
11	5.989133	b6:17:48:34:b4:ea	fe:5f:2a:19:53:33	ARP	42	10.0.0.2 is at b6:17:48:34:b4:ea
12	5.990017	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0x0003, seq=5
13	5.990053	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply id=0x0003, seq=5

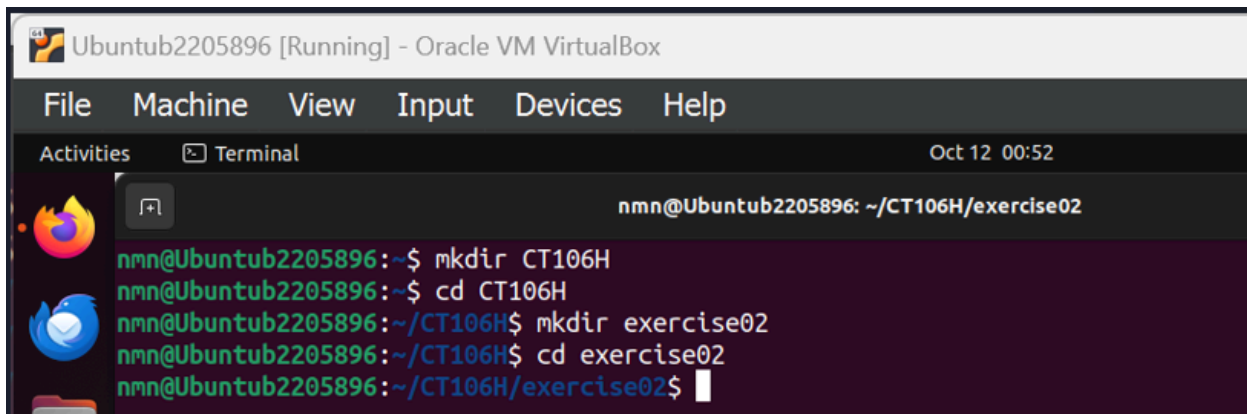
Exercise 2: Construct the following network.

Step 1 - create a folder named CT106H and exercise02 inside CT106H

Answer:

```
$ mkdir CT106H ; cd CT106H/ CT106H
```

```
$ mkdir exercise02 ; $ cd exercise02
```



The screenshot shows a terminal window titled 'Ubuntub2205896 [Running] - Oracle VM VirtualBox'. The terminal has a menu bar with 'File', 'Machine', 'View', 'Input', 'Devices', and 'Help'. Below the menu bar, there's a status bar with 'Activities', 'Terminal', and the date 'Oct 12 00:52'. The terminal prompt is 'nmn@Ubuntub2205896: ~/CT106H/exercise02'. The terminal output shows the following commands and their results:

```
nmn@Ubuntub2205896:~$ mkdir CT106H
nmn@Ubuntub2205896:~$ cd CT106H
nmn@Ubuntub2205896:~/CT106H$ mkdir exercise02
nmn@Ubuntub2205896:~/CT106H$ cd exercise02
nmn@Ubuntub2205896:~/CT106H/exercise02$
```

Step 2 - create the folders and files in tree inside exercise02

Answer:

```
$ mkdir pc{1..4}
```

```
$ mkdir router1
```

```
$ mkdir shared
```

```
$ gedit lab.conf
```

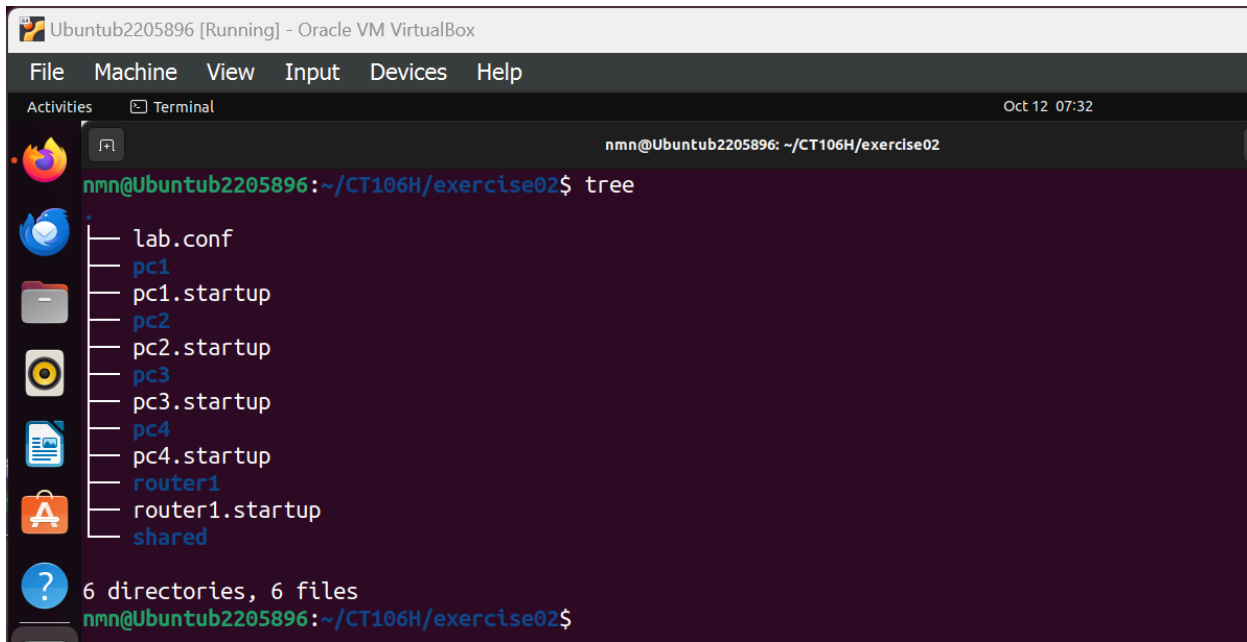
```
$ gedit pc1.startup
```

```
$ gedit pc2.startup
```

```
$ gedit pc3.startup
```

```
$ gedit pc4.startup
```

```
$ gedit router1.startup
```



The screenshot shows a terminal window titled 'Ubuntub2205896 [Running] - Oracle VM VirtualBox'. The terminal has a menu bar with 'File', 'Machine', 'View', 'Input', 'Devices', and 'Help'. Below the menu bar, there's a status bar with 'Activities', 'Terminal', and the date 'Oct 12 07:32'. The terminal prompt is 'nmn@Ubuntub2205896: ~/CT106H/exercise02'. The terminal output shows the following command and its result:

```
nmn@Ubuntub2205896:~/CT106H/exercise02$ tree
.
├── lab.conf
├── pc1
├── pc1.startup
├── pc2
├── pc2.startup
├── pc3
├── pc3.startup
├── pc4
├── pc4.startup
├── router1
├── router1.startup
└── shared

6 directories, 6 files
nmn@Ubuntub2205896:~/CT106H/exercise02$
```

```

nmn@Ubuntu2205896: ~/CT106H/exercise02
nmn@Ubuntu2205896:~/CT106H/exercise02$ cat lab.conf
pc1[0]=A
pc3[0]=A
pc2[0]=B
pc4[0]=B
router1[0]=A
router1[1]=B
nmn@Ubuntu2205896:~/CT106H/exercise02$ cat pc1.startup
ifconfig eth0 10.0.0.101/24 up
route add default gw 10.0.0.1
nmn@Ubuntu2205896:~/CT106H/exercise02$ cat pc2.startup
ifconfig eth0 10.0.1.101/24 up
route add default gw 10.0.1.1
nmn@Ubuntu2205896:~/CT106H/exercise02$ cat pc3.startup
ifconfig eth0 10.0.0.102/24 up
route add default gw 10.0.0.1
nmn@Ubuntu2205896:~/CT106H/exercise02$ cat pc4.startup
ifconfig eth0 10.0.1.102/24 up
route add default gw 10.0.1.1
nmn@Ubuntu2205896:~/CT106H/exercise02$ cat router1.startup
ifconfig eth0 10.0.0.1/24 up
ifconfig eth1 10.0.1.1/24 up
nmn@Ubuntu2205896:~/CT106H/exercise02$

```

Step 3 - running these commands

Answer:

\$ kathara lstart

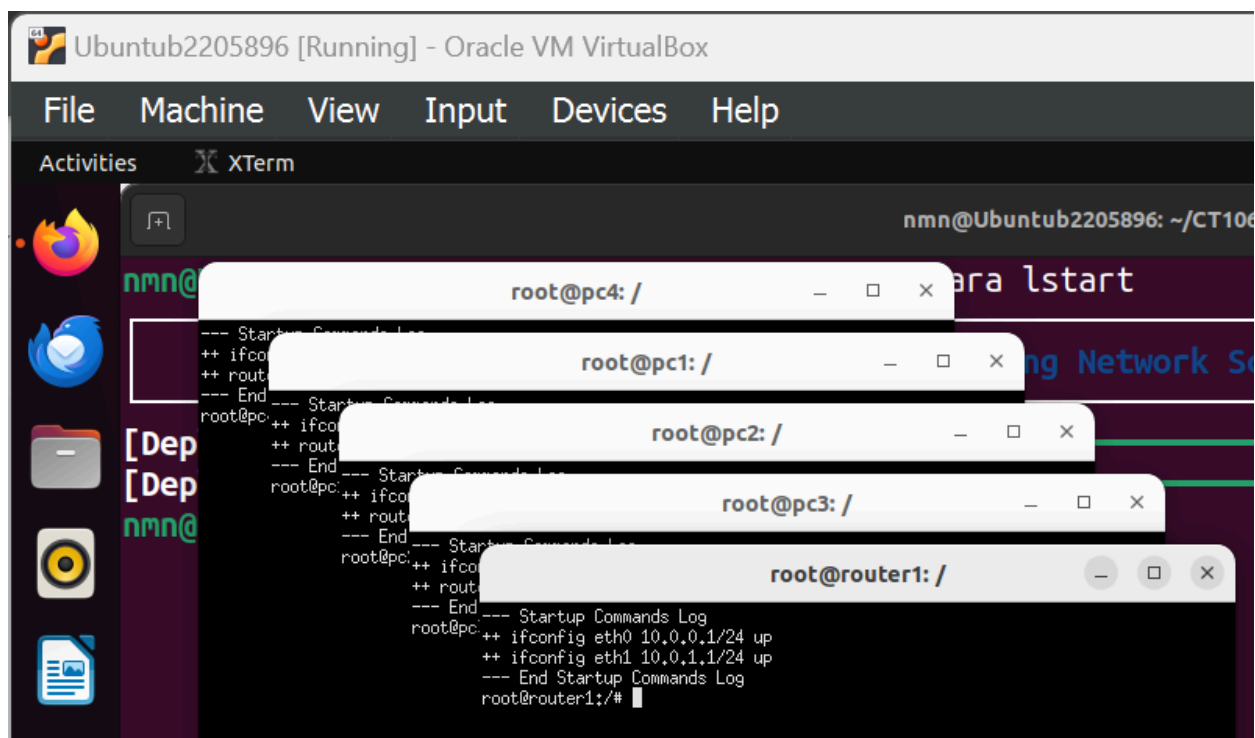
\$ kathara lclean

```

nmn@Ubuntu2205896: ~/CT106H/exercise02
nmn@Ubuntu2205896:~/CT106H/exercise02$ kathara lstart
Starting Network Scenario
[Deploying collision domains] 2/2
[Deploying devices] 5/5
nmn@Ubuntu2205896:~/CT106H/exercise02$ kathara lclean
Stopping Network Scenario
[Deleting devices] 5/5
[Deleting collision domains] 2/2
nmn@Ubuntu2205896:~/CT106H/exercise02$

```

Result:



Exercise 3: Construct the following network:

Step 1 - create new folder exercise03 and inside files

Answer:

```
$ mkdir exercise03 ; cd exercise03
```

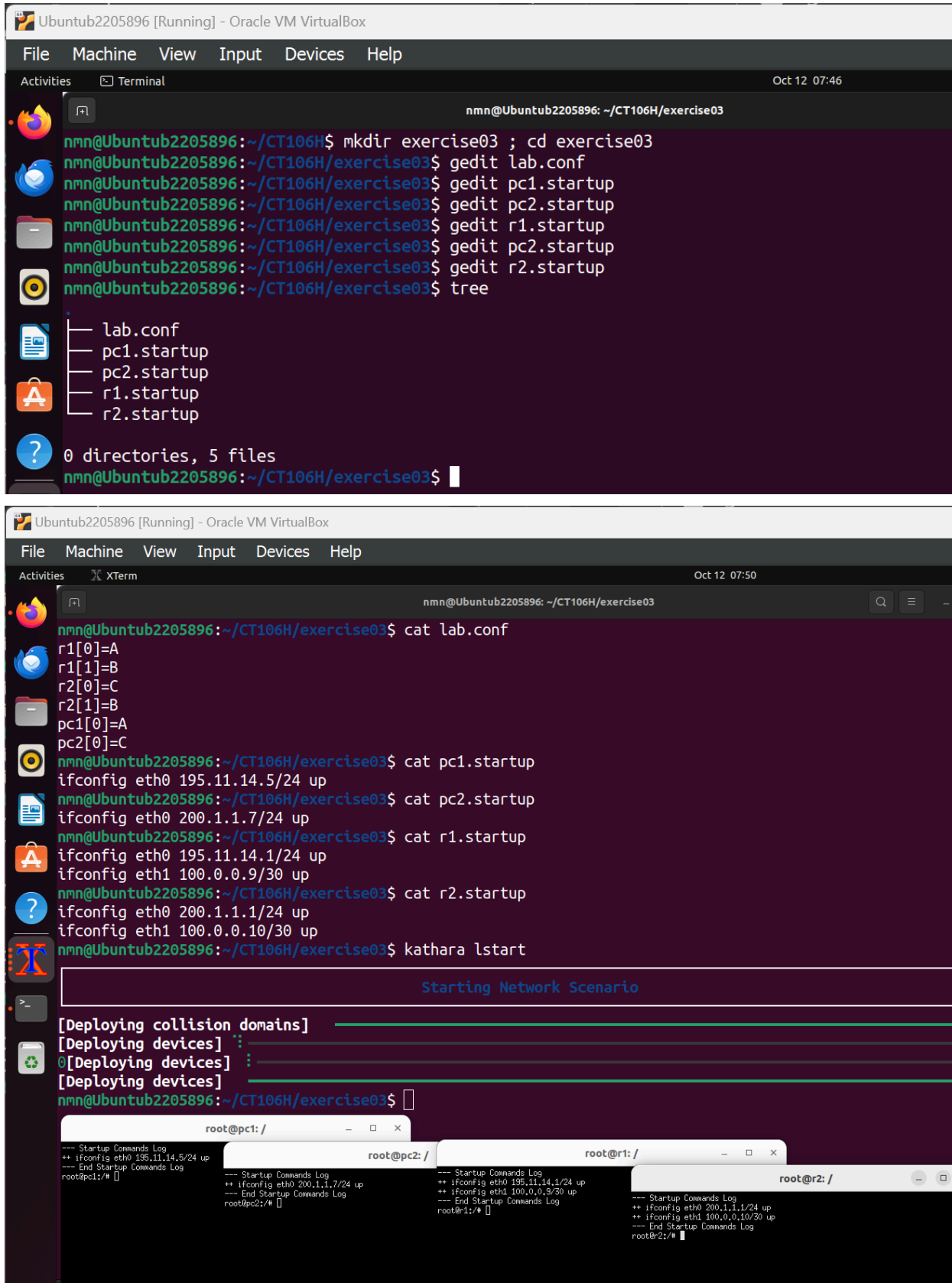
```
$ gedit lab.conf
```

```
$ gedit pc1.startup
```

```
$ gedit pc2.startup
```

```
$ gedit r1.startup
```

```
$ gedit r2.startup
```



```
nmn@Ubuntu2205896: ~/CT106H/exercise03
nmn@Ubuntu2205896:~/CT106H$ mkdir exercise03 ; cd exercise03
nmn@Ubuntu2205896:~/CT106H/exercise03$ gedit lab.conf
nmn@Ubuntu2205896:~/CT106H/exercise03$ gedit pc1.startup
nmn@Ubuntu2205896:~/CT106H/exercise03$ gedit pc2.startup
nmn@Ubuntu2205896:~/CT106H/exercise03$ gedit r1.startup
nmn@Ubuntu2205896:~/CT106H/exercise03$ gedit pc2.startup
nmn@Ubuntu2205896:~/CT106H/exercise03$ gedit r2.startup
nmn@Ubuntu2205896:~/CT106H/exercise03$ tree
.
├── lab.conf
├── pc1.startup
├── pc2.startup
├── r1.startup
└── r2.startup

0 directories, 5 files
nmn@Ubuntu2205896:~/CT106H/exercise03$
```

```
nmn@Ubuntu2205896:~/CT106H/exercise03$ cat lab.conf
r1[0]=A
r1[1]=B
r2[0]=C
r2[1]=B
pc1[0]=A
pc2[0]=C
nmn@Ubuntu2205896:~/CT106H/exercise03$ cat pc1.startup
ifconfig eth0 195.11.14.5/24 up
nmn@Ubuntu2205896:~/CT106H/exercise03$ cat pc2.startup
ifconfig eth0 200.1.1.7/24 up
nmn@Ubuntu2205896:~/CT106H/exercise03$ cat r1.startup
ifconfig eth0 195.11.14.1/24 up
ifconfig eth1 100.0.0.9/30 up
nmn@Ubuntu2205896:~/CT106H/exercise03$ cat r2.startup
ifconfig eth0 200.1.1.1/24 up
ifconfig eth1 100.0.0.10/30 up
nmn@Ubuntu2205896:~/CT106H/exercise03$ kathara lstart

Starting Network Scenario

[Deploying collision domains]
[Deploying devices]
0[Deploying devices]
[Deploying devices]
nmn@Ubuntu2205896:~/CT106H/exercise03$
```

```
root@pc1: /
--- Startup Commands Log
++ ifconfig eth0 195.11.14.5/24 up
--- End Startup Commands Log
root@pc1: /#

root@pc2: /
--- Startup Commands Log
++ ifconfig eth0 200.1.1.7/24 up
--- End Startup Commands Log
root@pc2: /#

root@r1: /
--- Startup Commands Log
++ ifconfig eth0 195.11.14.1/24 up
++ ifconfig eth1 100.0.0.9/30 up
--- End Startup Commands Log
root@r1: /#

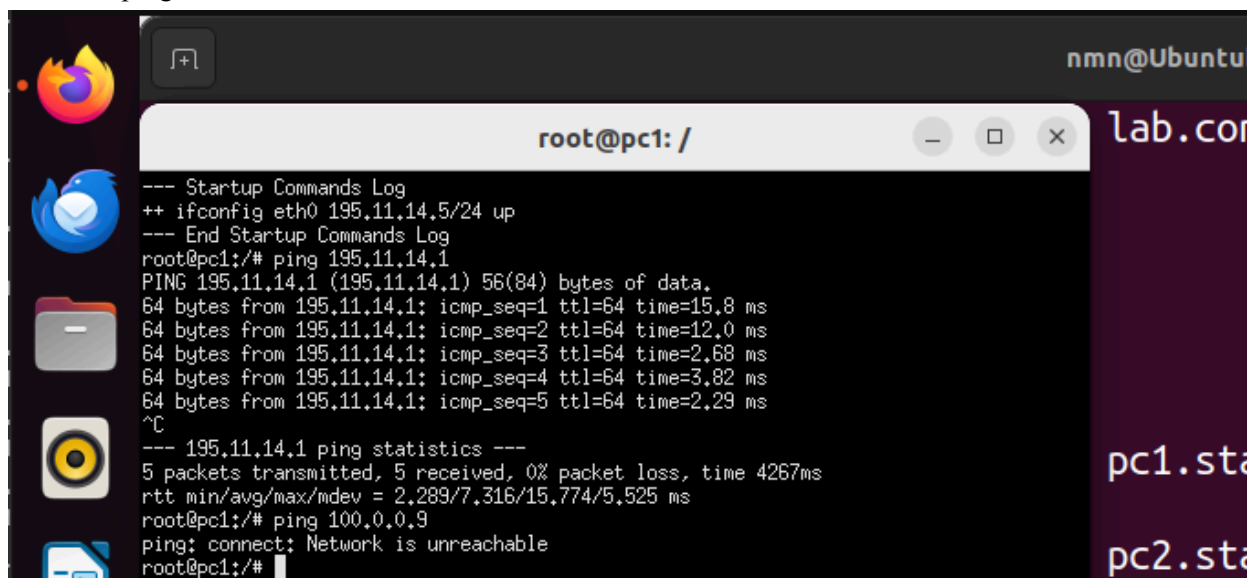
root@r2: /
--- Startup Commands Log
++ ifconfig eth0 200.1.1.1/24 up
++ ifconfig eth1 100.0.0.10/30 up
--- End Startup Commands Log
root@r2: /#
```

Step 2 - testing connectivity

Answer:

(In pc1) # ping 195.11.14.1

ping 100.0.0.9



```

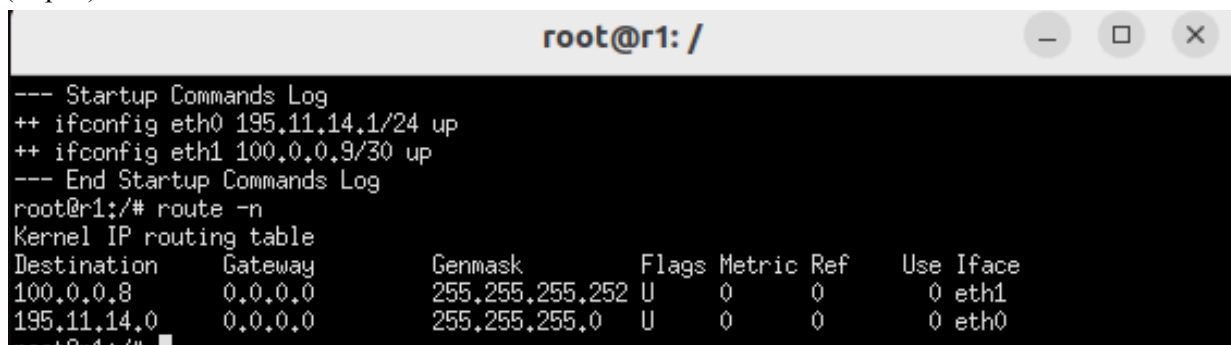
root@pc1: /
--- Startup Commands Log
++ ifconfig eth0 195.11.14.5/24 up
--- End Startup Commands Log
root@pc1:/# ping 195.11.14.1
PING 195.11.14.1 (195.11.14.1) 56(84) bytes of data:
64 bytes from 195.11.14.1: icmp_seq=1 ttl=64 time=15.8 ms
64 bytes from 195.11.14.1: icmp_seq=2 ttl=64 time=12.0 ms
64 bytes from 195.11.14.1: icmp_seq=3 ttl=64 time=2.68 ms
64 bytes from 195.11.14.1: icmp_seq=4 ttl=64 time=3.82 ms
64 bytes from 195.11.14.1: icmp_seq=5 ttl=64 time=2.29 ms
^C
--- 195.11.14.1 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4267ms
rtt min/avg/max/mdev = 2.289/7.316/15.774/5.525 ms
root@pc1:/# ping 100.0.0.9
ping: connect: Network is unreachable
root@pc1:/#

```

Step 3 - inspecting routing tables:

Answer:

(In pc1) # route -n



```

root@r1: /
--- Startup Commands Log
++ ifconfig eth0 195.11.14.1/24 up
++ ifconfig eth1 100.0.0.9/30 up
--- End Startup Commands Log
root@r1:/# route -n
Kernel IP routing table
Destination    Gateway         Genmask         Flags Metric Ref    Use Iface
100.0.0.0      0.0.0.0         255.255.255.252 U         0      0        0 eth1
195.11.14.0    0.0.0.0         255.255.255.0   U         0      0        0 eth0
root@r1:/#

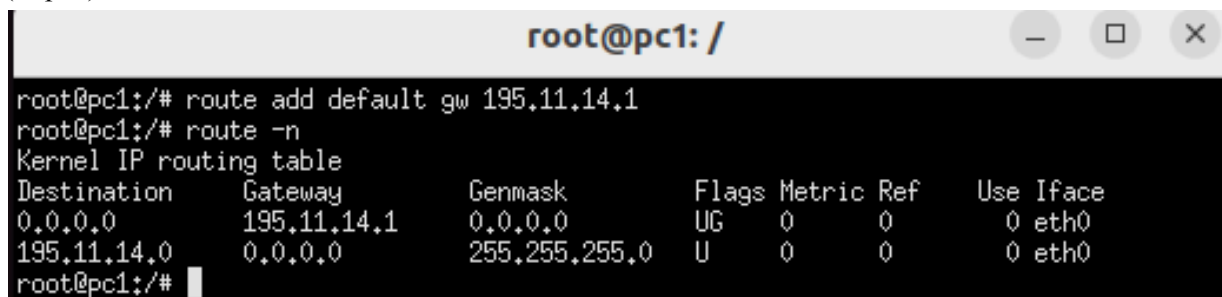
```

Step 4 - default routes on PCs:

Answer:

(In pc1) # route add default gw 195.11.14.1

(In pc1) # route -n



```

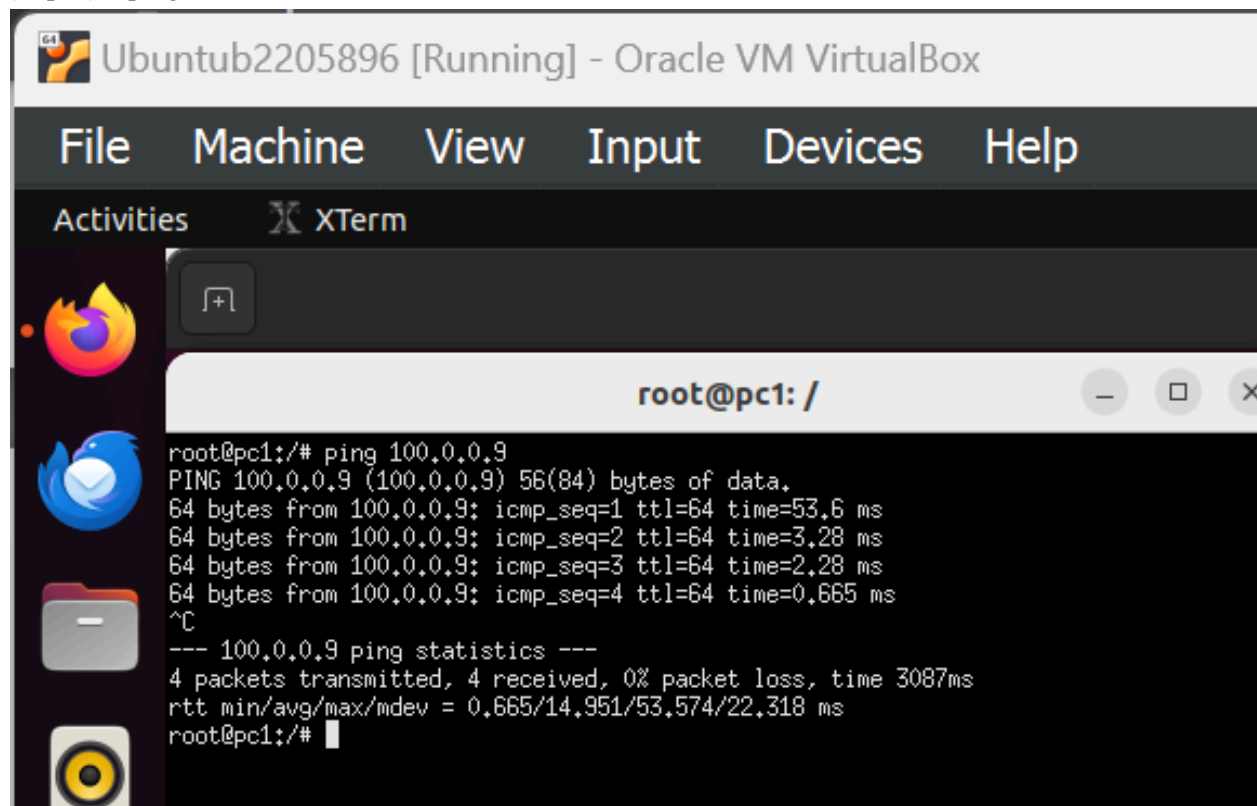
root@pc1: /
root@pc1:/# route add default gw 195.11.14.1
root@pc1:/# route -n
Kernel IP routing table
Destination    Gateway         Genmask         Flags Metric Ref    Use Iface
0.0.0.0        195.11.14.1    0.0.0.0         UG        0      0        0 eth0
195.11.14.0    0.0.0.0         255.255.255.0   U         0      0        0 eth0
root@pc1:/#

```

Step 4 - default routes on PCs: test

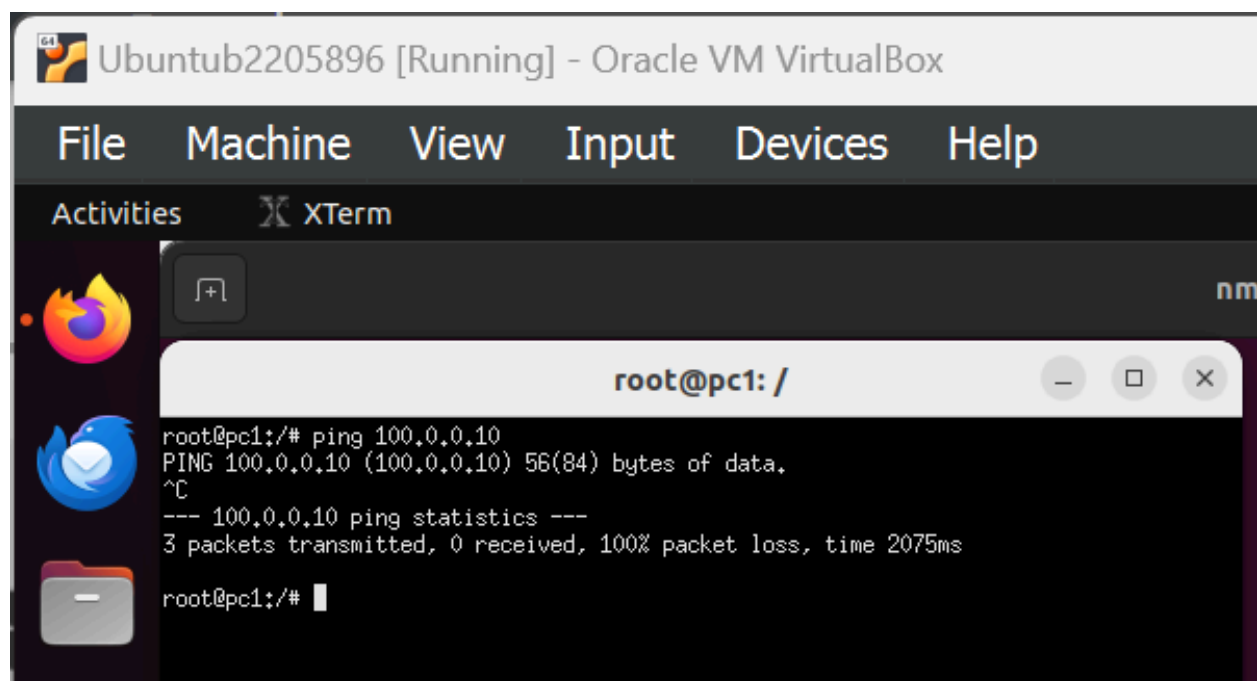
Answer:

(In pc1) # ping 100.0.0.9



The screenshot shows a terminal window titled "root@pc1: /" within the Oracle VM VirtualBox environment. The terminal output shows a successful ping to 100.0.0.9. The ping command was executed, and the output shows 4 packets transmitted, 4 received, 0% packet loss, and a time of 3087ms. The terminal window is part of a larger interface with a menu bar (File, Machine, View, Input, Devices, Help) and a sidebar with icons for Activities, XTerm, and other applications.

```
root@pc1:/# ping 100.0.0.9
PING 100.0.0.9 (100.0.0.9) 56(84) bytes of data:
64 bytes from 100.0.0.9: icmp_seq=1 ttl=64 time=53.6 ms
64 bytes from 100.0.0.9: icmp_seq=2 ttl=64 time=3.28 ms
64 bytes from 100.0.0.9: icmp_seq=3 ttl=64 time=2.28 ms
64 bytes from 100.0.0.9: icmp_seq=4 ttl=64 time=0.665 ms
^C
--- 100.0.0.9 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3087ms
rtt min/avg/max/mdev = 0.665/14.951/53.574/22.318 ms
root@pc1:/#
```



The screenshot shows a terminal window titled "root@pc1: /" within the Oracle VM VirtualBox environment. The terminal output shows a failed ping to 100.0.0.10. The ping command was executed, and the output shows 3 packets transmitted, 0 received, 100% packet loss, and a time of 2075ms. The terminal window is part of a larger interface with a menu bar (File, Machine, View, Input, Devices, Help) and a sidebar with icons for Activities, XTerm, and other applications.

```
root@pc1:/# ping 100.0.0.10
PING 100.0.0.10 (100.0.0.10) 56(84) bytes of data:
^C
--- 100.0.0.10 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2075ms
root@pc1:/#
```

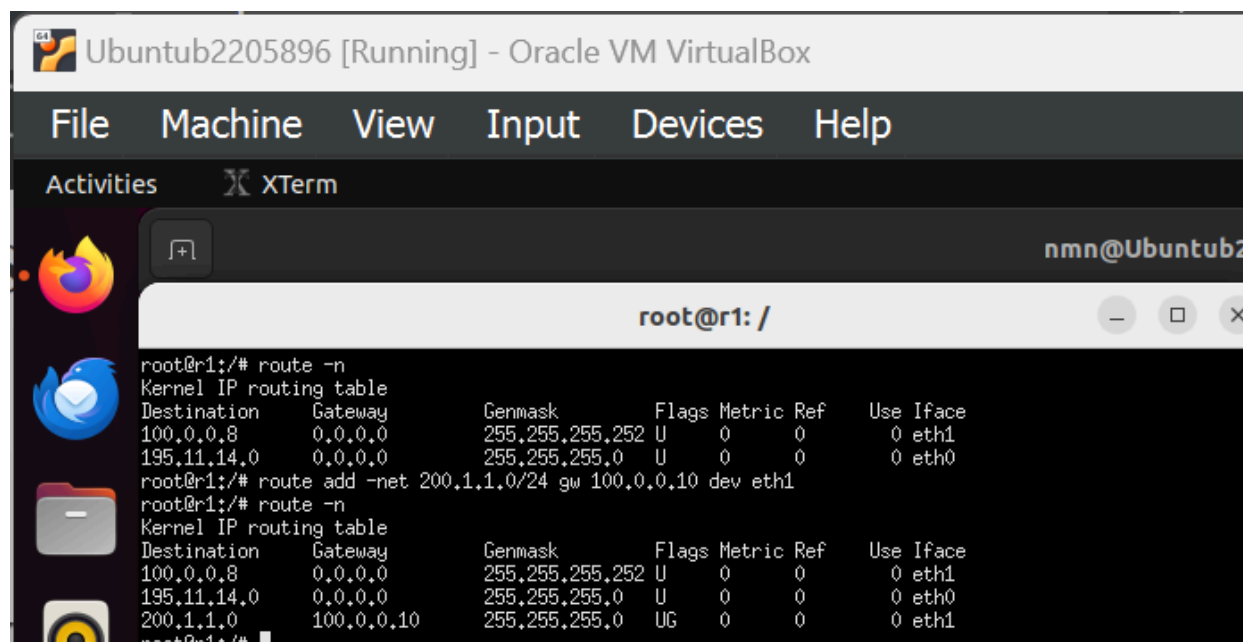
Step 6: Configuring a static route for r1, r2, pc2:

Answer:

(In r1) # route -n

route add -net 200.1.1.0/24 gw 100.0.0.10 dev eth1

route -n



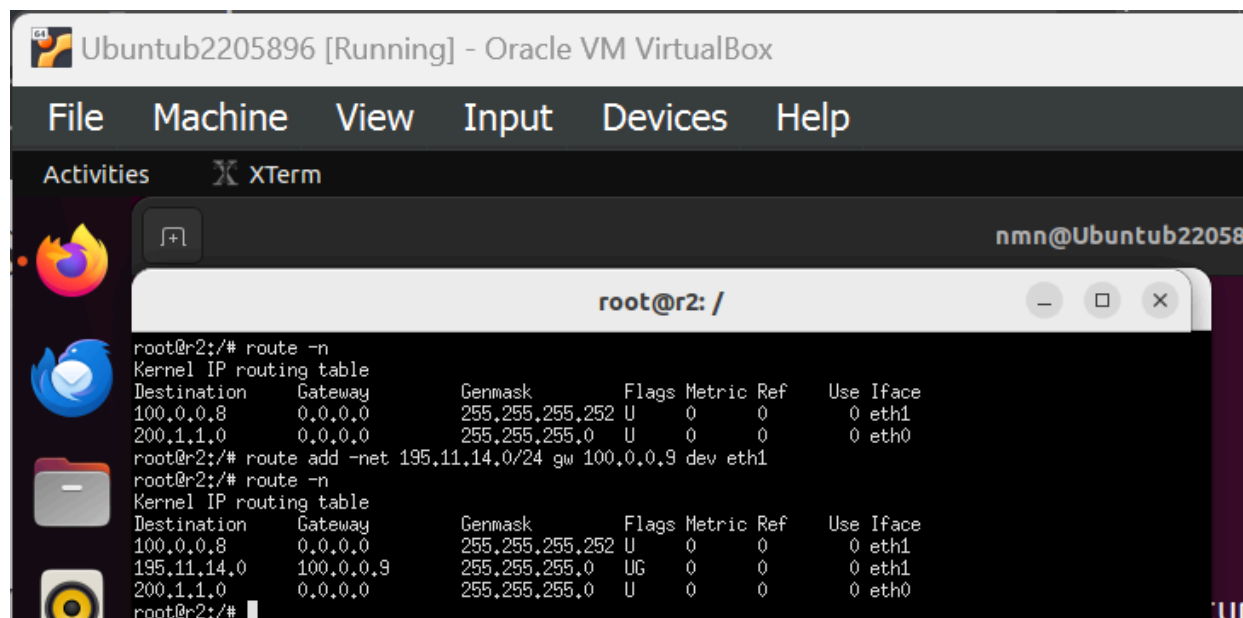
The screenshot shows a terminal window titled 'root@r1: /' within an Ubuntu VM. The user has executed the command 'route -n', which displays the current kernel IP routing table. The table lists two entries: a destination of 100.0.0.8 with gateway 0.0.0.0 and interface eth1, and a destination of 195.11.14.0 with gateway 0.0.0.0 and interface eth0. The user then executes 'route add -net 200.1.1.0/24 gw 100.0.0.10 dev eth1'. After running 'route -n' again, the routing table now includes a third entry for destination 200.1.1.0 with gateway 100.0.0.10 and interface eth1.

```
root@r1: /  
root@r1:~# route -n  
Kernel IP routing table  
Destination Gateway Genmask Flags Metric Ref Use Iface  
100.0.0.8 0.0.0.0 255.255.255.252 U 0 0 0 eth1  
195.11.14.0 0.0.0.0 255.255.255.0 U 0 0 0 eth0  
root@r1:~# route add -net 200.1.1.0/24 gw 100.0.0.10 dev eth1  
root@r1:~# route -n  
Kernel IP routing table  
Destination Gateway Genmask Flags Metric Ref Use Iface  
100.0.0.8 0.0.0.0 255.255.255.252 U 0 0 0 eth1  
195.11.14.0 0.0.0.0 255.255.255.0 U 0 0 0 eth0  
200.1.1.0 100.0.0.10 255.255.255.0 UG 0 0 0 eth1  
root@r1:~#
```

(In r2) # route -n

route add -net 195.11.14.0/24 gw 100.0.0.9 dev eth1

route -n



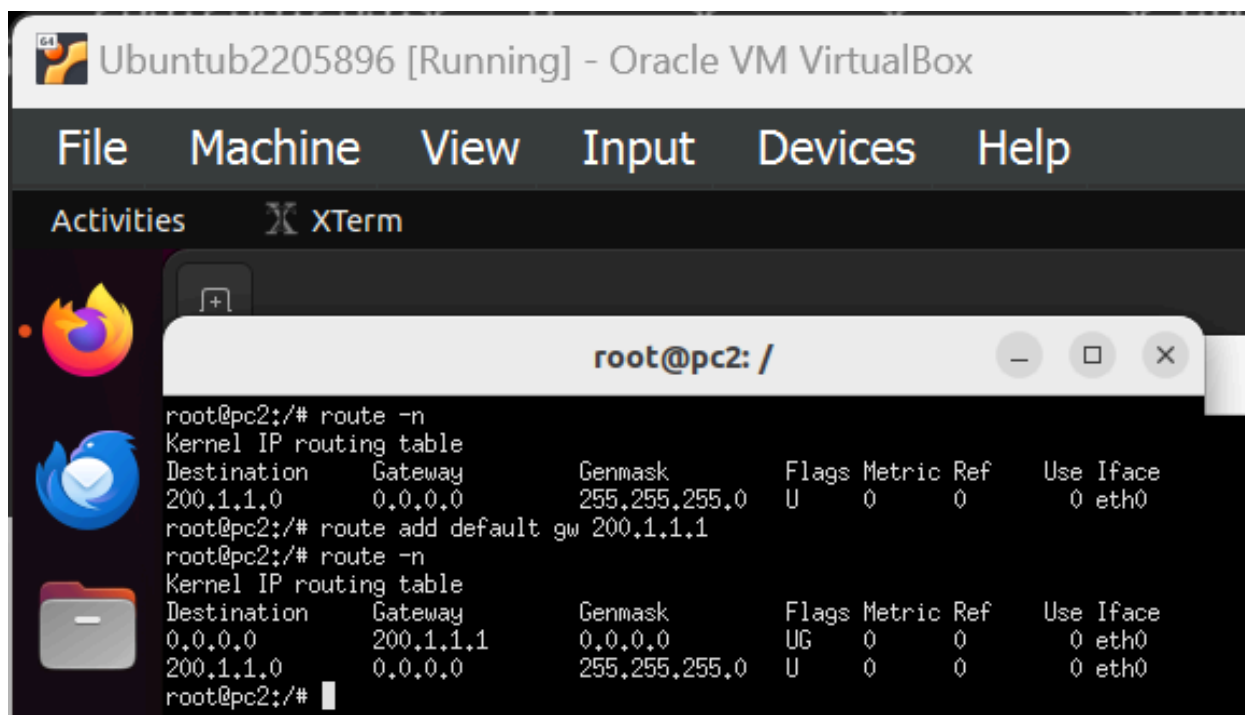
The screenshot shows a terminal window titled 'root@r2: /' within an Ubuntu VM. The user has executed the command 'route -n', which displays the current kernel IP routing table. The table lists two entries: a destination of 100.0.0.8 with gateway 0.0.0.0 and interface eth1, and a destination of 200.1.1.0 with gateway 0.0.0.0 and interface eth0. The user then executes 'route add -net 195.11.14.0/24 gw 100.0.0.9 dev eth1'. After running 'route -n' again, the routing table now includes a third entry for destination 195.11.14.0 with gateway 100.0.0.9 and interface eth1.

```
root@r2: /  
root@r2:~# route -n  
Kernel IP routing table  
Destination Gateway Genmask Flags Metric Ref Use Iface  
100.0.0.8 0.0.0.0 255.255.255.252 U 0 0 0 eth1  
200.1.1.0 0.0.0.0 255.255.255.0 U 0 0 0 eth0  
root@r2:~# route add -net 195.11.14.0/24 gw 100.0.0.9 dev eth1  
root@r2:~# route -n  
Kernel IP routing table  
Destination Gateway Genmask Flags Metric Ref Use Iface  
100.0.0.8 0.0.0.0 255.255.255.252 U 0 0 0 eth1  
195.11.14.0 100.0.0.9 255.255.255.0 UG 0 0 0 eth1  
200.1.1.0 0.0.0.0 255.255.255.0 U 0 0 0 eth0  
root@r2:~#
```

(In pc2) # route -n

route add default gw 200.1.1.1 dev eth0


```
# route -n
```

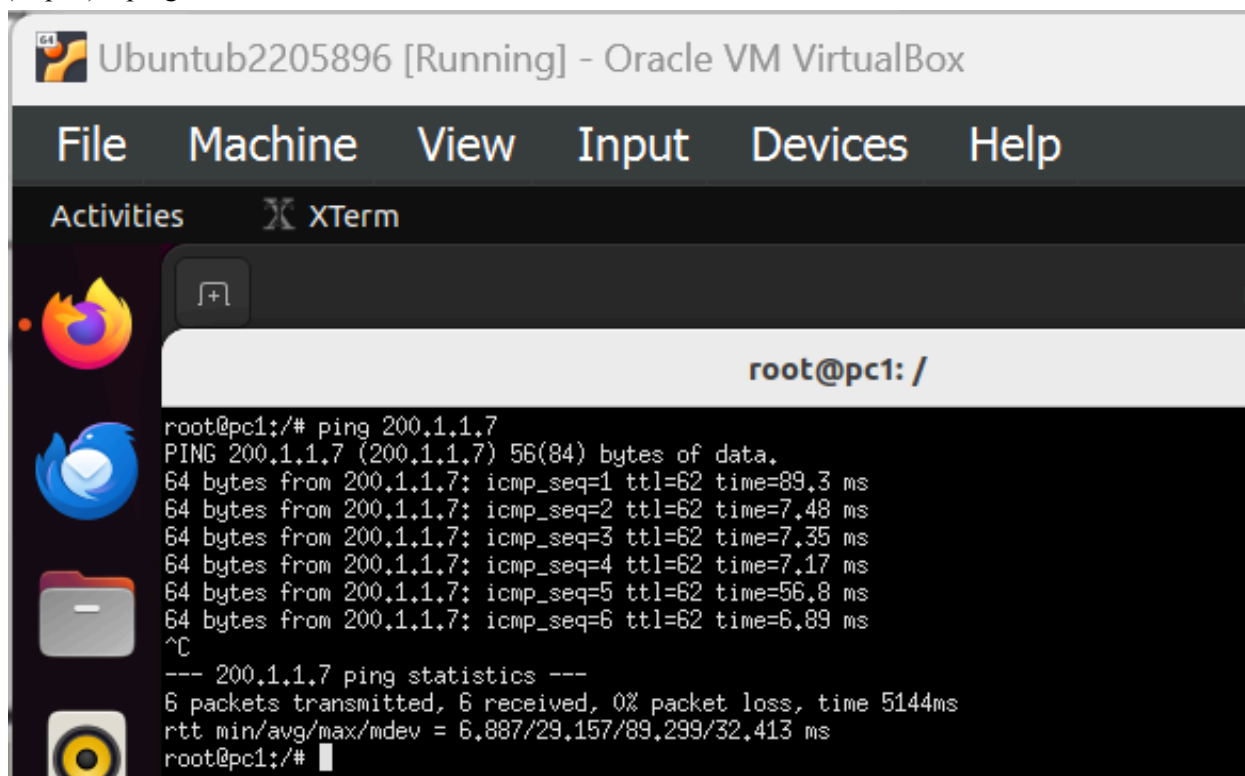


```
root@pc2:/# route -n
Kernel IP routing table
Destination Gateway Genmask Flags Metric Ref Use Iface
200.1.1.0 0.0.0.0 255.255.255.0 U 0 0 0 eth0
root@pc2:/# route add default gw 200.1.1.1
root@pc2:/# route -n
Kernel IP routing table
Destination Gateway Genmask Flags Metric Ref Use Iface
0.0.0.0 200.1.1.1 0.0.0.0 UG 0 0 0 eth0
200.1.1.0 0.0.0.0 255.255.255.0 U 0 0 0 eth0
root@pc2:/#
```

Step 7 - testing static routers:

Answer:

(In pc1) # ping 200.1.1.7



```
root@pc1:/# ping 200.1.1.7
PING 200.1.1.7 (200.1.1.7) 56(84) bytes of data:
64 bytes from 200.1.1.7: icmp_seq=1 ttl=62 time=89.3 ms
64 bytes from 200.1.1.7: icmp_seq=2 ttl=62 time=7.48 ms
64 bytes from 200.1.1.7: icmp_seq=3 ttl=62 time=7.35 ms
64 bytes from 200.1.1.7: icmp_seq=4 ttl=62 time=7.17 ms
64 bytes from 200.1.1.7: icmp_seq=5 ttl=62 time=56.8 ms
64 bytes from 200.1.1.7: icmp_seq=6 ttl=62 time=6.89 ms
^C
--- 200.1.1.7 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5144ms
rtt min/avg/max/mdev = 6.887/29.157/89.299/32.413 ms
root@pc1:/#
```

Exercise 4: Study arp protocol

Answer:

Step 1 - create a new folder exercise04 and inside files

```
$ mkdir exercise04 ; cd exercise04
```

```
$ gedit lab.conf
```

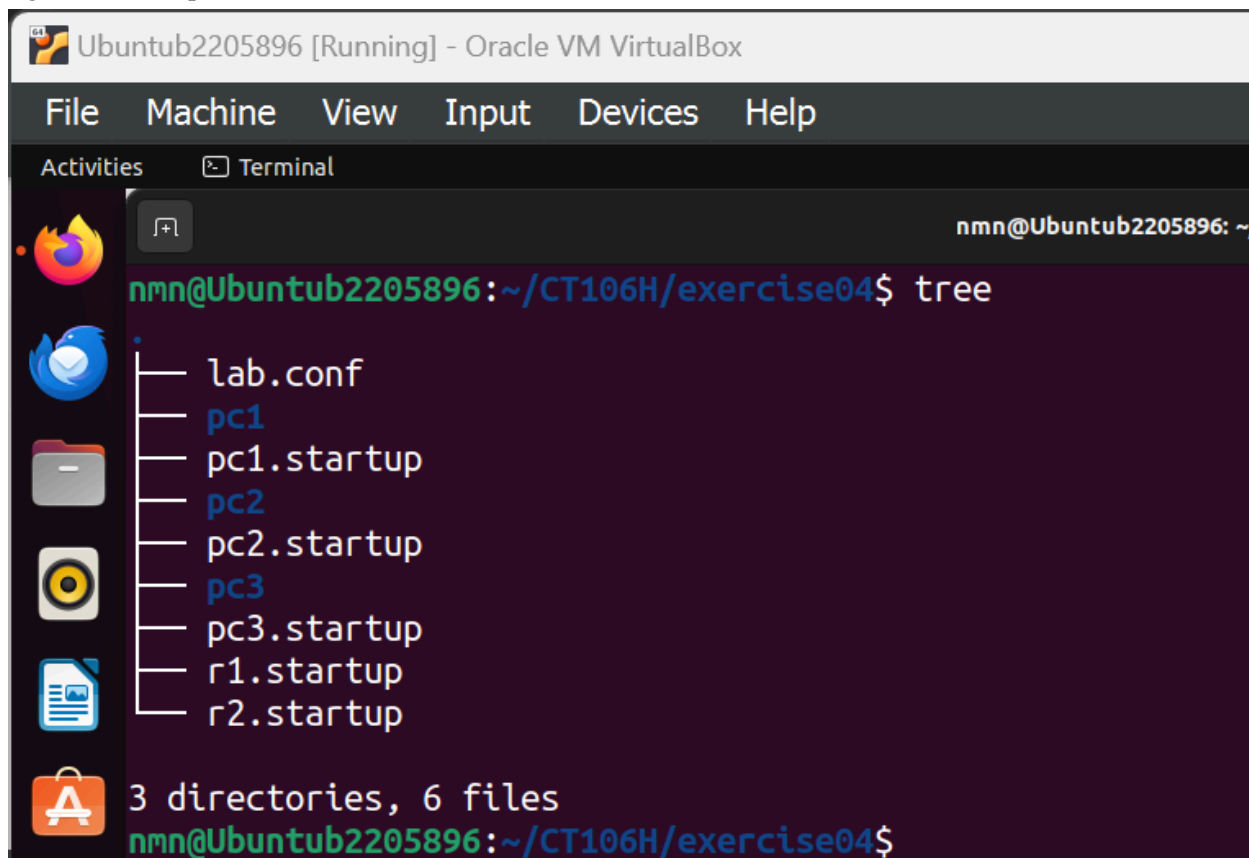
```
$ gedit pc1.startup
```

```
$ gedit pc2.startup
```

```
$ gedit pc3.startup
```

```
$ gedit r1.startup
```

```
$ gedit r2.startup
```



The screenshot shows a terminal window titled "Ubuntu2205896 [Running] - Oracle VM VirtualBox". The terminal displays the output of the 'tree' command in the directory ~/CT106H/exercise04. The output shows a directory structure with 3 directories and 6 files. The files are lab.conf, pc1, pc2, pc3, pc1.startup, pc2.startup, pc3.startup, r1.startup, and r2.startup. The terminal prompt is nm@Ubuntu2205896: ~/CT106H/exercise04\$.

```
nm@Ubuntu2205896:~/CT106H/exercise04$ tree
.
├── lab.conf
├── pc1
├── pc1.startup
├── pc2
├── pc2.startup
├── pc3
├── pc3.startup
├── r1.startup
└── r2.startup

3 directories, 6 files
nm@Ubuntu2205896:~/CT106H/exercise04$
```

```

nmn@Ubuntub2205896: ~/CT106H/exercise04
nmn@Ubuntub2205896:~/CT106H/exercise04$ cat lab.conf
r1[0]="A"
r1[1]="B"
r2[0]="C"
r2[1]="B"
pc1[0]="A"
pc2[0]="C"
pc3[0]="C"
nmn@Ubuntub2205896:~/CT106H/exercise04$ cat pc1.startup
ifconfig eth0 195.11.14.5 up
route add default gw 195.11.14.1
nmn@Ubuntub2205896:~/CT106H/exercise04$ cat pc2.startup
ifconfig eth0 200.1.1.7 up
route add default gw 200.1.1.1
nmn@Ubuntub2205896:~/CT106H/exercise04$ cat pc3.startup
ifconfig eth0 200.1.1.3 up
route add default gw 200.1.1.1
nmn@Ubuntub2205896:~/CT106H/exercise04$ cat r1.startup
ifconfig eth0 195.11.14.1 up
ifconfig eth1 100.0.0.9 netmask 255.255.255.252 broadcast 100.0.0.11 up
route add -net 200.1.1.0 netmask 255.255.255.0 gw 100.0.0.10 dev eth1
nmn@Ubuntub2205896:~/CT106H/exercise04$ cat r2.startup
ifconfig eth0 200.1.1.1 up
ifconfig eth1 100.0.0.10 netmask 255.255.255.252 broadcast 100.0.0.11 up
route add -net 195.11.14.0 netmask 255.255.255.0 gw 100.0.0.9 dev eth1
nmn@Ubuntub2205896:~/CT106H/exercise04$

```

Step 2 - start the lab:

Answer:

\$ kathara lstart

```

nmn@Ubuntub2205896: ~/CT106H/exercise04
nmn@Ubuntub2205896:~/CT106H/exercise04$ kathara lstart

Starting Network Scenario

[Deploying collision domains]
[Deploying devices]
nmn@Ubuntub2205896:~/CT106H/exercise04$

root@pc2: /
--- Startup
++ ifconfig
++ route add
--- End Start--- Startup Command Log
root@pc2:/# ++ ifconfig eth0 200.1.1.7 up
++ route add default gw 200.1.1.1
--- End Start--- Startup Command Log
root@pc1:/# ++ ifconfig eth0 195.11.14.5 up
++ ifconfig eth1 100.0.0.9 netmask 255.255.255.252 broadcast 100.0.0.11 up
++ route add -net 200.1.1.0 netmask 255.255.255.0 gw 100.0.0.10 dev eth1
--- End Start--- Startup Command Log
root@r2:/# ++ ifconfig eth0 200.1.1.1 up
++ ifconfig eth1 100.0.0.10 netmask 255.255.255.252 broadcast 100.0.0.11 up
++ route add -net 195.11.14.0 netmask 255.255.255.0 gw 100.0.0.9 dev eth1
--- End Start--- Startup Command Log
root@r1:/# ++ ifconfig eth0 195.11.14.1 up
++ ifconfig eth1 100.0.0.9 netmask 255.255.255.252 broadcast 100.0.0.11 up
++ route add -net 200.1.1.0 netmask 255.255.255.0 gw 100.0.0.10 dev eth1
--- End Start--- Startup Command Log
root@pc3:/# ++ ifconfig eth0 200.1.1.3 up
++ route add default gw 200.1.1.1
--- End Start--- Startup Command Log

```

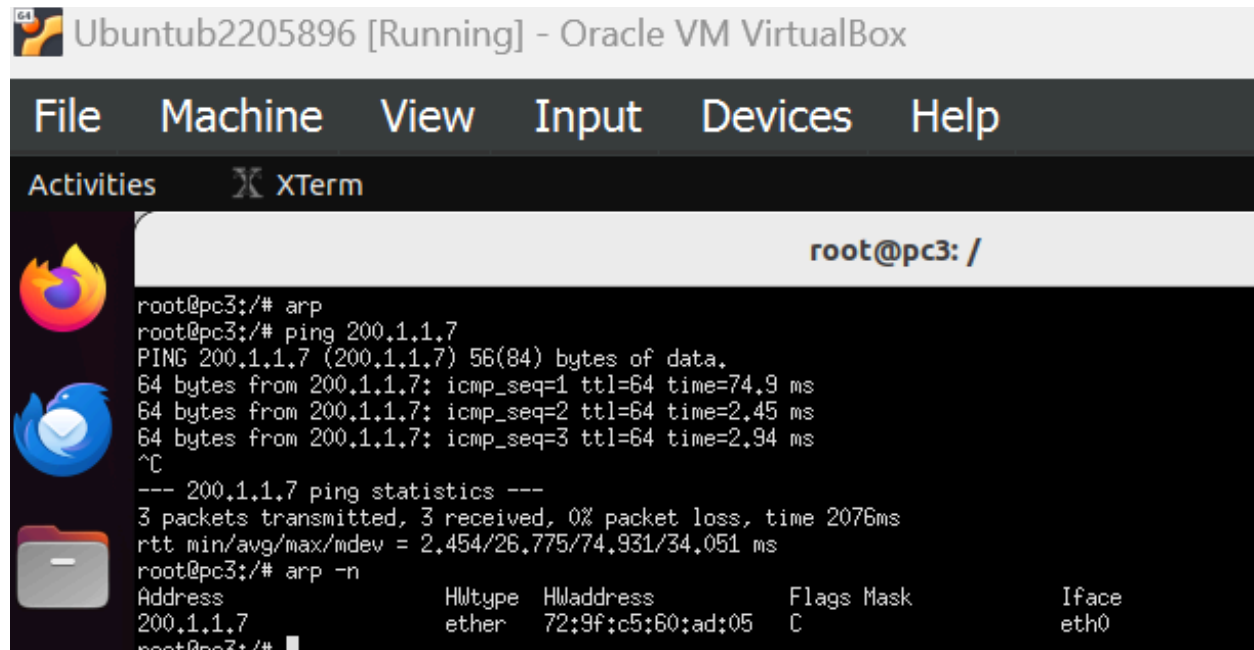
Step 3 - inspecting the arp cache (local traffic):

Answer:

(In pc3) # arp

ping 200.1.1.7

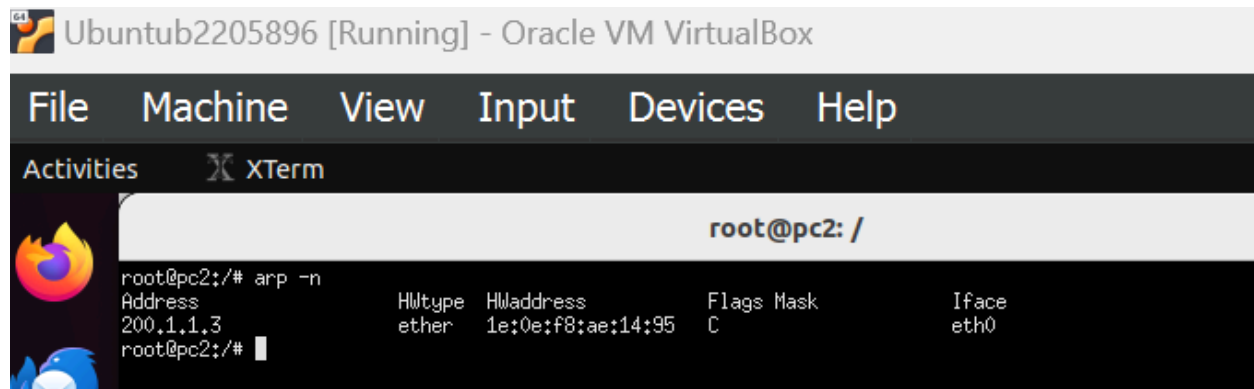
arp -n



```

root@pc3:/# arp
root@pc3:/# ping 200.1.1.7
PING 200.1.1.7 (200.1.1.7) 56(84) bytes of data:
64 bytes from 200.1.1.7: icmp_seq=1 ttl=64 time=74.9 ms
64 bytes from 200.1.1.7: icmp_seq=2 ttl=64 time=2.45 ms
64 bytes from 200.1.1.7: icmp_seq=3 ttl=64 time=2.94 ms
^C
--- 200.1.1.7 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2076ms
rtt min/avg/max/mdev = 2.454/26.775/74.931/34.051 ms
root@pc3:/# arp -n
Address                  Hwtype  Hwaddress      Flags Mask          Iface
200.1.1.7                ether   72:9f:c5:60:ad:05 C                    eth0
root@pc3:/#
  
```

(In pc2) # arp -n



```

root@pc2:/# arp -n
Address                  Hwtype  Hwaddress      Flags Mask          Iface
200.1.1.3                ether   1e:0e:f8:ae:14:95 C                    eth0
root@pc2:/#
  
```

Step 4: inspecting the arp cache (non local traffic)

Answer:

(In pc2) # ping 195.11.14.5

arp -n

```

root@pc2: /
root@pc2:/# ping 195.11.14.5
PING 195.11.14.5 (195.11.14.5) 56(84) bytes of data.
64 bytes from 195.11.14.5: icmp_seq=1 ttl=62 time=126 ms
64 bytes from 195.11.14.5: icmp_seq=2 ttl=62 time=7.33 ms
^C
--- 195.11.14.5 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1170ms
rtt min/avg/max/mdev = 7.331/66.888/126.446/59.557 ms
root@pc2:/# arp -n
Address                  Hwtype  Hwaddress      Flags Mask          Iface
200.1.1.3                ether   1e:0e:f8:ae:14:95 C                    eth0
200.1.1.1                ether   0a:ab:4d:8b:c5:47 C                    eth0
root@pc2:/#

```

(In router 1 and 2)

arp -n

```

root@r1: /
root@r1:/# arp -n
Address                  Hwtype  Hwaddress      Flags Mask          Iface
195.11.14.5             ether   82:37:f7:fc:d2:90 C                    eth0
100.0.0.10              ether   4e:88:98:f8:e6:f8 C                    eth1
root@r1:/#

```

```

root@r2: /
root@r2:/# arp -n
Address                  Hwtype  Hwaddress      Flags Mask          Iface
100.0.0.9               ether   ae:62:d1:80:a8:67 C                    eth1
200.1.1.7               ether   72:9f:c5:60:ad:05 C                    eth0
root@r2:/#

```

Step 5 - sniffing arp traffic: Restart the lab in order to clear arp caches:

\$ kathara lclean

\$ kathara lstart

```

nmn@Ubuntu2205896: ~/CT106H/exercise04
nmn@Ubuntu2205896:~/CT106H/exercise04$ kathara lclean

Stopping Network Scenario

[Deleting devices]
[Deleting collision domains]
nmn@Ubuntu2205896:~/CT106H/exercise04$ kathara lstart

Starting Network Scenario

[Deploying collision domains]
[Deploying devices]
nmn@Ubuntu2205896:~/CT106H/exercise04$

```

Step 6

(In pc2) # ping 195.11.14.5

```

root@pc2: /
root@pc2: /# ping 195.11.14.5
PING 195.11.14.5 (195.11.14.5) 56(84) bytes of data:
64 bytes from 195.11.14.5: icmp_seq=1 ttl=62 time=37.0 ms
64 bytes from 195.11.14.5: icmp_seq=2 ttl=62 time=39.3 ms
64 bytes from 195.11.14.5: icmp_seq=3 ttl=62 time=9.26 ms
64 bytes from 195.11.14.5: icmp_seq=4 ttl=62 time=7.40 ms
64 bytes from 195.11.14.5: icmp_seq=5 ttl=62 time=7.20 ms
64 bytes from 195.11.14.5: icmp_seq=6 ttl=62 time=15.9 ms
64 bytes from 195.11.14.5: icmp_seq=7 ttl=62 time=9.05 ms
^C
--- 195.11.14.5 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6058ms
rtt min/avg/max/mdev = 7.201/17.869/39.291/13.109 ms
root@pc2: /#

```

While running the ping in pc2, run this command in r2:

tcpdump -tnni eth0

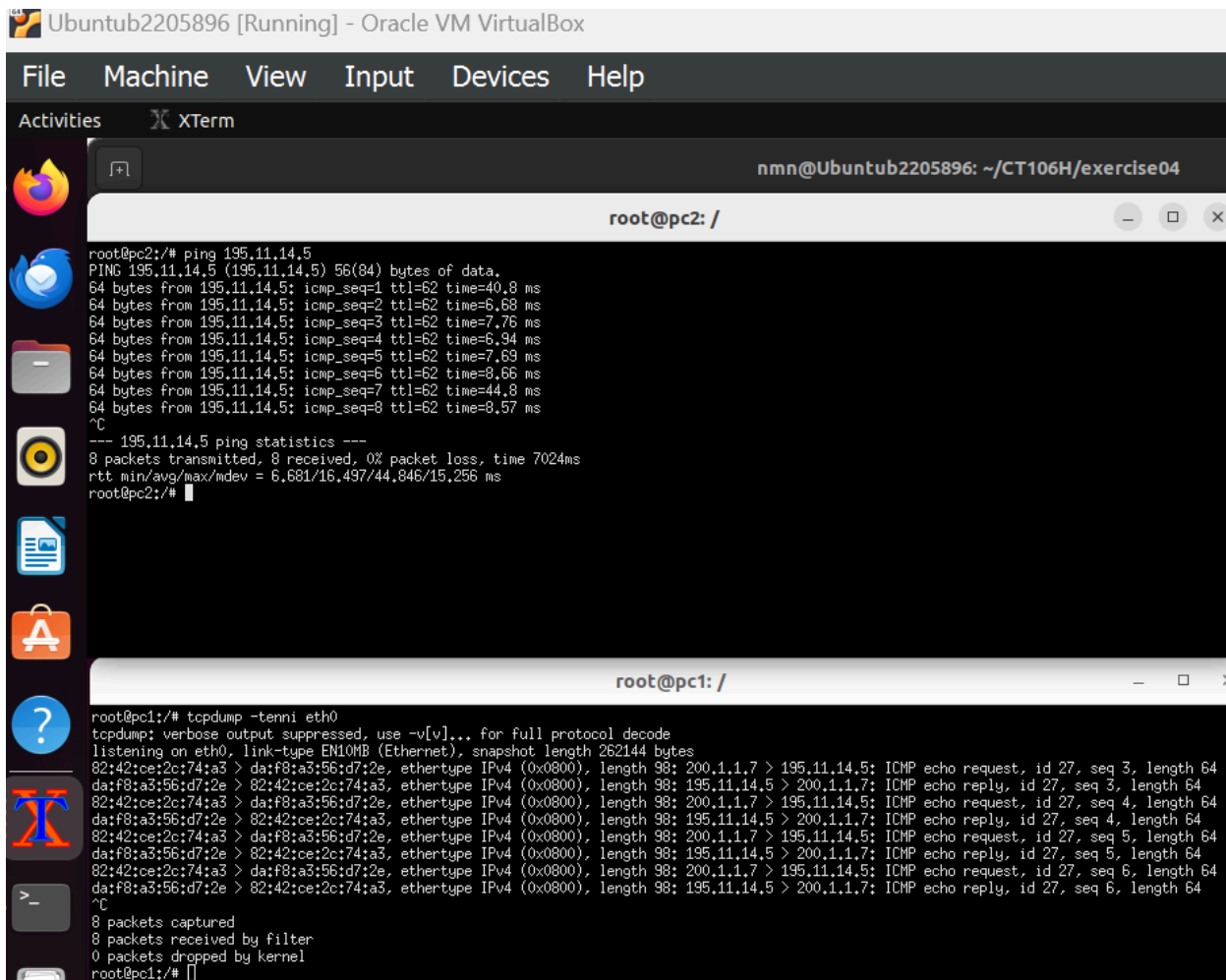
```

root@r2: /
root@r2: /# tcpdump -tnni eth0
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
ee:21:86:92:88:32 > 0a:2e:64:6b:d5:78, ethertype IPv4 (0x0800), length 98: 200.1.1.7 > 195.11.14.5: ICMP echo request, id 28, seq 3, length 64
0a:2e:64:6b:d5:78 > ee:21:86:92:88:32, ethertype IPv4 (0x0800), length 98: 195.11.14.5 > 200.1.1.7: ICMP echo reply, id 28, seq 3, length 64
ee:21:86:92:88:32 > 0a:2e:64:6b:d5:78, ethertype IPv4 (0x0800), length 98: 200.1.1.7 > 195.11.14.5: ICMP echo request, id 28, seq 4, length 64
0a:2e:64:6b:d5:78 > ee:21:86:92:88:32, ethertype IPv4 (0x0800), length 98: 195.11.14.5 > 200.1.1.7: ICMP echo reply, id 28, seq 4, length 64
ee:21:86:92:88:32 > 0a:2e:64:6b:d5:78, ethertype IPv4 (0x0800), length 98: 200.1.1.7 > 195.11.14.5: ICMP echo request, id 28, seq 5, length 64
0a:2e:64:6b:d5:78 > ee:21:86:92:88:32, ethertype IPv4 (0x0800), length 98: 195.11.14.5 > 200.1.1.7: ICMP echo reply, id 28, seq 5, length 64
ee:21:86:92:88:32 > 0a:2e:64:6b:d5:78, ethertype IPv4 (0x0800), length 98: 200.1.1.7 > 195.11.14.5: ICMP echo request, id 28, seq 6, length 64
0a:2e:64:6b:d5:78 > ee:21:86:92:88:32, ethertype IPv4 (0x0800), length 98: 195.11.14.5 > 200.1.1.7: ICMP echo reply, id 28, seq 6, length 64
0a:2e:64:6b:d5:78 > ee:21:86:92:88:32, ethertype ARP (0x0806), length 42: Request who-has 200.1.1.7 tell 200.1.1.1, length 28
ee:21:86:92:88:32 > 0a:2e:64:6b:d5:78, ethertype ARP (0x0806), length 60: Request who-has 200.1.1.1 tell 200.1.1.7, length 46
0a:2e:64:6b:d5:78 > ee:21:86:92:88:32, ethertype ARP (0x0806), length 42: Reply 200.1.1.1 is-at 0a:2e:64:6b:d5:78, length 28
ee:21:86:92:88:32 > 0a:2e:64:6b:d5:78, ethertype ARP (0x0806), length 60: Reply 200.1.1.7 is-at ee:21:86:92:88:32, length 46
^C
12 packets captured
12 packets received by filter
0 packets dropped by kernel
root@r2: /#

```

While running the ping in pc2, run this command in r1:

tcpdump -tnni eth1



```
nmn@Ubuntub2205896: ~/CT106H/exercise04

root@pc2: /

root@pc2:~# ping 195.11.14.5
PING 195.11.14.5 (195.11.14.5) 56(84) bytes of data.
64 bytes from 195.11.14.5: icmp_seq=1 ttl=62 time=40.8 ms
64 bytes from 195.11.14.5: icmp_seq=2 ttl=62 time=6.68 ms
64 bytes from 195.11.14.5: icmp_seq=3 ttl=62 time=7.76 ms
64 bytes from 195.11.14.5: icmp_seq=4 ttl=62 time=6.94 ms
64 bytes from 195.11.14.5: icmp_seq=5 ttl=62 time=7.69 ms
64 bytes from 195.11.14.5: icmp_seq=6 ttl=62 time=8.66 ms
64 bytes from 195.11.14.5: icmp_seq=7 ttl=62 time=44.8 ms
64 bytes from 195.11.14.5: icmp_seq=8 ttl=62 time=8.57 ms
^C
--- 195.11.14.5 ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 7024ms
rtt min/avg/max/mdev = 6.681/16.497/44.846/15.256 ms
root@pc2:~#

root@pc1: /

root@pc1:~# tcpdump -tnni eth0
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
82:42:ce:2c:74:a3 > da:f8:a3:56:d7:2e, ethertype IPv4 (0x0800), length 98: 200.1.1.7 > 195.11.14.5: ICMP echo request, id 27, seq 3, length 64
da:f8:a3:56:d7:2e > 82:42:ce:2c:74:a3, ethertype IPv4 (0x0800), length 98: 195.11.14.5 > 200.1.1.7: ICMP echo reply, id 27, seq 3, length 64
82:42:ce:2c:74:a3 > da:f8:a3:56:d7:2e, ethertype IPv4 (0x0800), length 98: 200.1.1.7 > 195.11.14.5: ICMP echo request, id 27, seq 4, length 64
da:f8:a3:56:d7:2e > 82:42:ce:2c:74:a3, ethertype IPv4 (0x0800), length 98: 195.11.14.5 > 200.1.1.7: ICMP echo reply, id 27, seq 4, length 64
82:42:ce:2c:74:a3 > da:f8:a3:56:d7:2e, ethertype IPv4 (0x0800), length 98: 200.1.1.7 > 195.11.14.5: ICMP echo request, id 27, seq 5, length 64
da:f8:a3:56:d7:2e > 82:42:ce:2c:74:a3, ethertype IPv4 (0x0800), length 98: 195.11.14.5 > 200.1.1.7: ICMP echo reply, id 27, seq 5, length 64
82:42:ce:2c:74:a3 > da:f8:a3:56:d7:2e, ethertype IPv4 (0x0800), length 98: 200.1.1.7 > 195.11.14.5: ICMP echo request, id 27, seq 6, length 64
da:f8:a3:56:d7:2e > 82:42:ce:2c:74:a3, ethertype IPv4 (0x0800), length 98: 195.11.14.5 > 200.1.1.7: ICMP echo reply, id 27, seq 6, length 64
^C
8 packets captured
8 packets received by filter
0 packets dropped by kernel
root@pc1:~#
```

While running the ping in pc2, run this command in pc1:

```
# tcpdump -tnni eth0
```

```

nmn@Ubuntub2205896: ~/CT106H/exercise04

root@pc2: /

root@pc2:~# ping 195.11.14.5
PING 195.11.14.5 (195.11.14.5) 56(84) bytes of data.
64 bytes from 195.11.14.5: icmp_seq=1 ttl=62 time=40.8 ms
64 bytes from 195.11.14.5: icmp_seq=2 ttl=62 time=6.68 ms
64 bytes from 195.11.14.5: icmp_seq=3 ttl=62 time=7.76 ms
64 bytes from 195.11.14.5: icmp_seq=4 ttl=62 time=6.94 ms
64 bytes from 195.11.14.5: icmp_seq=5 ttl=62 time=7.69 ms
64 bytes from 195.11.14.5: icmp_seq=6 ttl=62 time=8.66 ms
64 bytes from 195.11.14.5: icmp_seq=7 ttl=62 time=44.8 ms
64 bytes from 195.11.14.5: icmp_seq=8 ttl=62 time=8.57 ms
^C
--- 195.11.14.5 ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 7024ms
rtt min/avg/max/mdev = 6.681/16.497/44.846/15.256 ms
root@pc2:~#

root@pc1: /

root@pc1:~# tcpdump -ttni eth0
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
82:42:ce:2c:74:a3 > da:f8:a3:56:d7:2e, ethertype IPv4 (0x0800), length 98: 200.1.1.7 > 195.11.14.5: ICMP echo request, id 27, seq 3, length 64
da:f8:a3:56:d7:2e > 82:42:ce:2c:74:a3, ethertype IPv4 (0x0800), length 98: 195.11.14.5 > 200.1.1.7: ICMP echo reply, id 27, seq 3, length 64
82:42:ce:2c:74:a3 > da:f8:a3:56:d7:2e, ethertype IPv4 (0x0800), length 98: 200.1.1.7 > 195.11.14.5: ICMP echo request, id 27, seq 4, length 64
da:f8:a3:56:d7:2e > 82:42:ce:2c:74:a3, ethertype IPv4 (0x0800), length 98: 195.11.14.5 > 200.1.1.7: ICMP echo reply, id 27, seq 4, length 64
82:42:ce:2c:74:a3 > da:f8:a3:56:d7:2e, ethertype IPv4 (0x0800), length 98: 200.1.1.7 > 195.11.14.5: ICMP echo request, id 27, seq 5, length 64
da:f8:a3:56:d7:2e > 82:42:ce:2c:74:a3, ethertype IPv4 (0x0800), length 98: 195.11.14.5 > 200.1.1.7: ICMP echo reply, id 27, seq 5, length 64
82:42:ce:2c:74:a3 > da:f8:a3:56:d7:2e, ethertype IPv4 (0x0800), length 98: 200.1.1.7 > 195.11.14.5: ICMP echo request, id 27, seq 6, length 64
da:f8:a3:56:d7:2e > 82:42:ce:2c:74:a3, ethertype IPv4 (0x0800), length 98: 195.11.14.5 > 200.1.1.7: ICMP echo reply, id 27, seq 6, length 64
^C
8 packets captured
8 packets received by filter
0 packets dropped by kernel
root@pc1:~#

```

Step 7 - arp implementation details: While running the ping in pc2, run this command in r2:

```
# tcpdump -ttni eth0
```


untub2205896 [Running] - Oracle VM VirtualBox

```

Machine View Input Devices Help
es XTerm
nmn@Ubuntu2205896: ~/CT106H/exercise04
root@pc2: /
root@pc2:/# ping 195.11.14.5
PING 195.11.14.5 (195.11.14.5) 56(84) bytes of data.
64 bytes from 195.11.14.5: icmp_seq=1 ttl=62 time=39.7 ms
64 bytes from 195.11.14.5: icmp_seq=2 ttl=62 time=7.09 ms
64 bytes from 195.11.14.5: icmp_seq=3 ttl=62 time=7.92 ms
64 bytes from 195.11.14.5: icmp_seq=4 ttl=62 time=8.95 ms
64 bytes from 195.11.14.5: icmp_seq=5 ttl=62 time=11.8 ms
64 bytes from 195.11.14.5: icmp_seq=6 ttl=62 time=8.20 ms
64 bytes from 195.11.14.5: icmp_seq=7 ttl=62 time=13.5 ms
64 bytes from 195.11.14.5: icmp_seq=8 ttl=62 time=5.21 ms
64 bytes from 195.11.14.5: icmp_seq=9 ttl=62 time=7.03 ms
^C
--- 195.11.14.5 ping statistics ---
9 packets transmitted, 9 received, 0% packet loss, time 8451ms
rtt min/avg/max/mdev = 5.209/12.158/39.738/10.037 ms
root@pc2:/#

root@r2: /
root@r2:/# tcpdump -t -i eth0
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
ee:21:86:92:88:32 > 0a:2e:64:6b:d5:78, ethertype IPv4 (0x0800), length 98: 200.1.1.7 > 195.11.14.5: ICMP echo request, id 30, seq 1, length 64
0a:2e:64:6b:d5:78 > ee:21:86:92:88:32, ethertype IPv4 (0x0800), length 98: 195.11.14.5 > 200.1.1.7: ICMP echo reply, id 30, seq 1, length 64
ee:21:86:92:88:32 > 0a:2e:64:6b:d5:78, ethertype IPv4 (0x0800), length 98: 200.1.1.7 > 195.11.14.5: ICMP echo request, id 30, seq 2, length 64
0a:2e:64:6b:d5:78 > ee:21:86:92:88:32, ethertype IPv4 (0x0800), length 98: 195.11.14.5 > 200.1.1.7: ICMP echo reply, id 30, seq 2, length 64
ee:21:86:92:88:32 > 0a:2e:64:6b:d5:78, ethertype IPv4 (0x0800), length 98: 200.1.1.7 > 195.11.14.5: ICMP echo request, id 30, seq 3, length 64
0a:2e:64:6b:d5:78 > ee:21:86:92:88:32, ethertype IPv4 (0x0800), length 98: 195.11.14.5 > 200.1.1.7: ICMP echo reply, id 30, seq 3, length 64
ee:21:86:92:88:32 > 0a:2e:64:6b:d5:78, ethertype IPv4 (0x0800), length 98: 200.1.1.7 > 195.11.14.5: ICMP echo request, id 30, seq 4, length 64
0a:2e:64:6b:d5:78 > ee:21:86:92:88:32, ethertype IPv4 (0x0800), length 98: 195.11.14.5 > 200.1.1.7: ICMP echo reply, id 30, seq 4, length 64
ee:21:86:92:88:32 > 0a:2e:64:6b:d5:78, ethertype IPv4 (0x0800), length 98: 200.1.1.7 > 195.11.14.5: ICMP echo request, id 30, seq 5, length 64
0a:2e:64:6b:d5:78 > ee:21:86:92:88:32, ethertype IPv4 (0x0800), length 98: 195.11.14.5 > 200.1.1.7: ICMP echo reply, id 30, seq 5, length 64
ee:21:86:92:88:32 > 0a:2e:64:6b:d5:78, ethertype IPv4 (0x0800), length 98: 200.1.1.7 > 195.11.14.5: ICMP echo request, id 30, seq 6, length 64
0a:2e:64:6b:d5:78 > ee:21:86:92:88:32, ethertype IPv4 (0x0800), length 98: 195.11.14.5 > 200.1.1.7: ICMP echo reply, id 30, seq 6, length 64
0a:2e:64:6b:d5:78 > ee:21:86:92:88:32, ethertype ARP (0x0806), length 42: Request who-has 200.1.1.7 tell 200.1.1.1, length 28
ee:21:86:92:88:32 > 0a:2e:64:6b:d5:78, ethertype ARP (0x0806), length 60: Reply 200.1.1.7 is-at ee:21:86:92:88:32, length 46
ee:21:86:92:88:32 > 0a:2e:64:6b:d5:78, ethertype ARP (0x0806), length 60: Request who-has 200.1.1.1 tell 200.1.1.7, length 46
0a:2e:64:6b:d5:78 > ee:21:86:92:88:32, ethertype ARP (0x0806), length 42: Reply 200.1.1.1 is-at 0a:2e:64:6b:d5:78, length 28
ee:21:86:92:88:32 > 0a:2e:64:6b:d5:78, ethertype IPv4 (0x0800), length 98: 200.1.1.7 > 195.11.14.5: ICMP echo request, id 30, seq 7, length 64
0a:2e:64:6b:d5:78 > ee:21:86:92:88:32, ethertype IPv4 (0x0800), length 98: 195.11.14.5 > 200.1.1.7: ICMP echo reply, id 30, seq 7, length 64
ee:21:86:92:88:32 > 0a:2e:64:6b:d5:78, ethertype IPv4 (0x0800), length 98: 200.1.1.7 > 195.11.14.5: ICMP echo request, id 30, seq 8, length 64
0a:2e:64:6b:d5:78 > ee:21:86:92:88:32, ethertype IPv4 (0x0800), length 98: 195.11.14.5 > 200.1.1.7: ICMP echo reply, id 30, seq 8, length 64
^C
20 packets captured
20 packets received by filter
0 packets dropped by kernel
root@r2:/#

```

Exercise 5: Construct the following network:

Step 1 - create a new folder exercise05 and inside files

Answer:

\$ mkdir exercise05 ; cd exercise05

\$ mkdir pc{1..3}

\$ gedit lab.conf

\$ gedit pc1.startup

\$ gedit pc2.startup

\$ gedit pc3.startup

\$ gedit router1.startup

\$ gedit router2.startup

\$ mkdir router1

\$ mkdir router2

\$ mkdir shared

Ubuntu2205896 [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

Activities Terminal Oct 12 10:...

nmn@Ubuntu2205896: ~/CT106H/exerices05

```
nmn@Ubuntu2205896:~/CT106H/exerices05$ tree
.
├── lab.conf
├── pc1
├── pc1.startup
├── pc2
├── pc2.startup
├── pc3
├── pc3.startup
├── router1
├── router1.startup
├── router2
├── router2.startup
└── shared

6 directories, 6 files
nmn@Ubuntu2205896:~/CT106H/exerices05$
```

Ubuntu2205896 [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

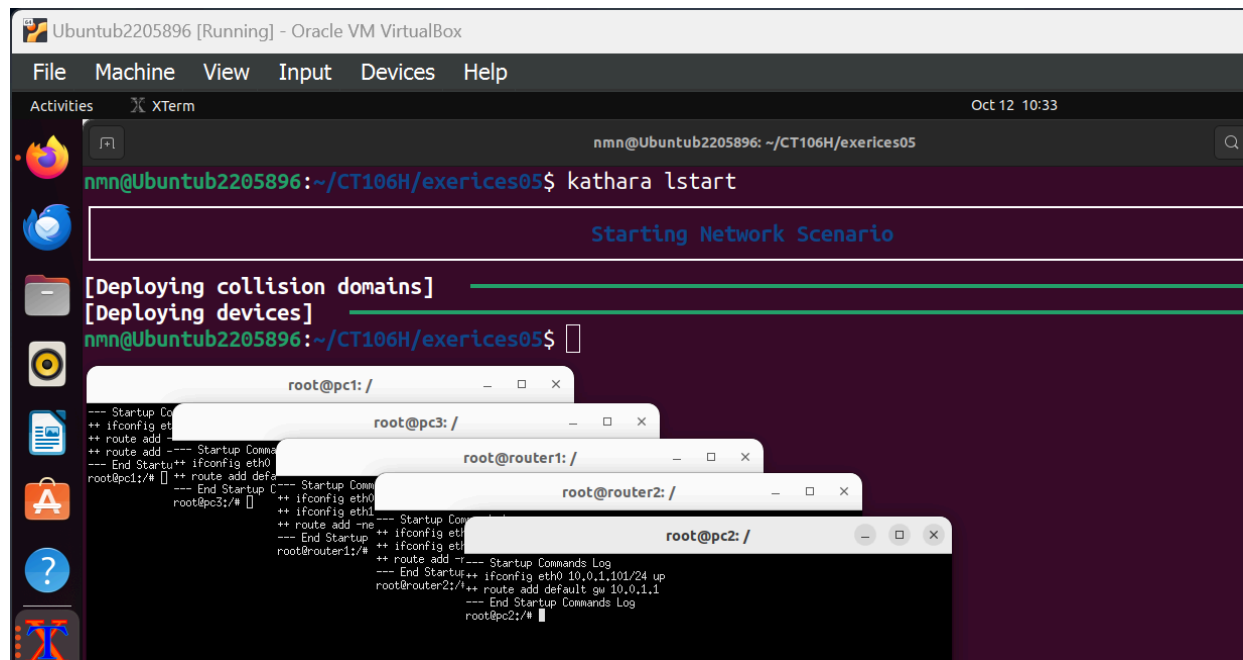
Activities Terminal Oct 12 10:27

nmn@Ubuntu2205896: ~/CT106H/exerices05

```
nmn@Ubuntu2205896:~/CT106H/exerices05$ ls
lab.conf  pc1.startup  pc2.startup  pc3.startup  router1.startup  router2.startup
pc1       pc2          pc3          router1      router2          shared
nmn@Ubuntu2205896:~/CT106H/exerices05$ cat lab.conf
pc1[0]=A
pc2[0]=B
pc3[0]=C
router1[0]=A
router1[1]=B
router2[0]=A
router2[1]=C
nmn@Ubuntu2205896:~/CT106H/exerices05$ cat router1.startup
ifconfig eth0 10.0.0.1/24 up
ifconfig eth1 10.0.1.1/24 up
route add -net 10.0.2.0/24 gw 10.0.0.2
nmn@Ubuntu2205896:~/CT106H/exerices05$ cat router2.startup
ifconfig eth0 10.0.0.2/24 up
ifconfig eth1 10.0.2.1/24 up
route add -net 10.0.1.0/24 gw 10.0.0.1
nmn@Ubuntu2205896:~/CT106H/exerices05$ cat pc1.startup
ifconfig eth0 10.0.0.101/24 up
route add -net 10.0.1.0/24 gw 10.0.0.1
route add -net 10.0.2.0/24 gw 10.0.0.2
nmn@Ubuntu2205896:~/CT106H/exerices05$ cat pc2.startup
ifconfig eth0 10.0.1.101/24 up
route add default gw 10.0.1.1
nmn@Ubuntu2205896:~/CT106H/exerices05$ cat pc3.startup
ifconfig eth0 10.0.2.101/24 up
route add default gw 10.0.2.1
nmn@Ubuntu2205896:~/CT106H/exerices05$
```

Step 2:

\$ kathara lstart

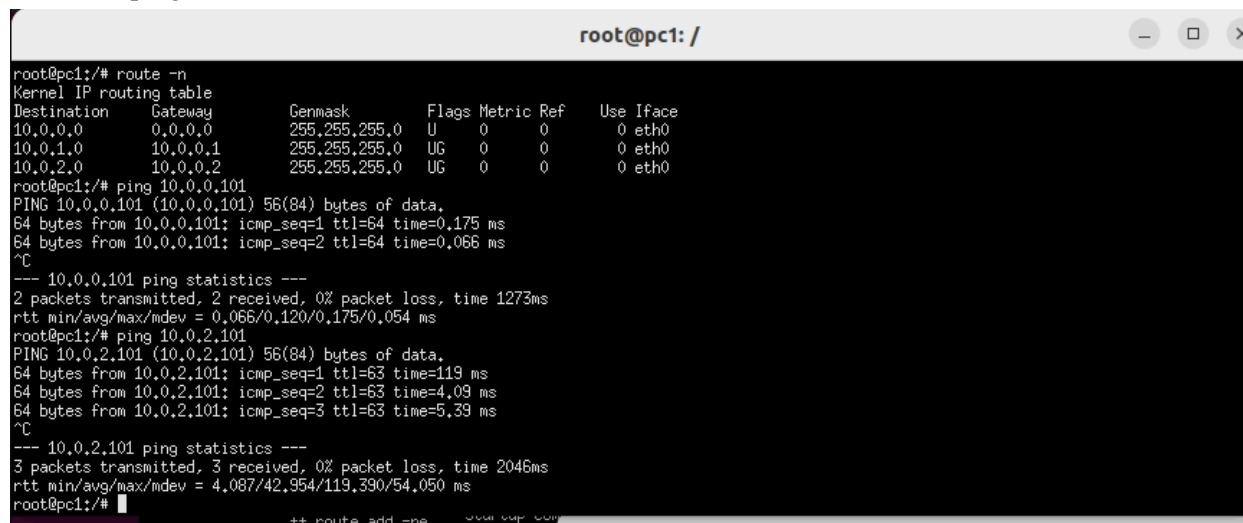


Step 3:

(In pc1) # route -n

ping 10.0.1.101

ping 10.0.2.101



(In pc2) # route -n

ping 10.0.2.101

```

root@pc2: /

--- Startup Commands Log
++ ifconfig eth0 10.0.1.101/24 up
++ route add default gw 10.0.1.1
--- End Startup Commands Log
root@pc2:/# route -n
Kernel IP routing table
Destination    Gateway         Genmask         Flags Metric Ref    Use Iface
0.0.0.0        10.0.1.1        0.0.0.0         UG    0      0      0 eth0
10.0.1.0       0.0.0.0         255.255.255.0   U     0      0      0 eth0
root@pc2:/# ping 10.0.2.101
PING 10.0.2.101 (10.0.2.101) 56(84) bytes of data.
64 bytes from 10.0.2.101: icmp_seq=1 ttl=62 time=121 ms
64 bytes from 10.0.2.101: icmp_seq=2 ttl=62 time=3.24 ms
64 bytes from 10.0.2.101: icmp_seq=3 ttl=62 time=3.47 ms
64 bytes from 10.0.2.101: icmp_seq=4 ttl=62 time=4.58 ms
^C
--- 10.0.2.101 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3350ms
rtt min/avg/max/mdev = 3.237/33.016/120.780/50.672 ms
root@pc2:/#

```

(In pc3) # route -n
ping 10.0.101

```

root@pc3: /

--- Startup Commands Log
++ ifconfig eth0 10.0.2.101/24 up
++ route add default gw 10.0.2.1
--- End Startup Commands Log
root@pc3:/# route -n
Kernel IP routing table
Destination    Gateway         Genmask         Flags Metric Ref    Use Iface
0.0.0.0        10.0.2.1        0.0.0.0         UG    0      0      0 eth0
10.0.2.0       0.0.0.0         255.255.255.0   U     0      0      0 eth0
root@pc3:/# ping 10.0.101
PING 10.0.101 (10.0.0.101) 56(84) bytes of data.
64 bytes from 10.0.0.101: icmp_seq=1 ttl=63 time=36.5 ms
64 bytes from 10.0.0.101: icmp_seq=2 ttl=63 time=4.55 ms
64 bytes from 10.0.0.101: icmp_seq=3 ttl=63 time=3.51 ms
64 bytes from 10.0.0.101: icmp_seq=4 ttl=63 time=5.16 ms
64 bytes from 10.0.0.101: icmp_seq=5 ttl=63 time=4.79 ms
^C
--- 10.0.101 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4054ms
rtt min/avg/max/mdev = 3.514/10.909/36.534/12.823 ms
root@pc3:/#

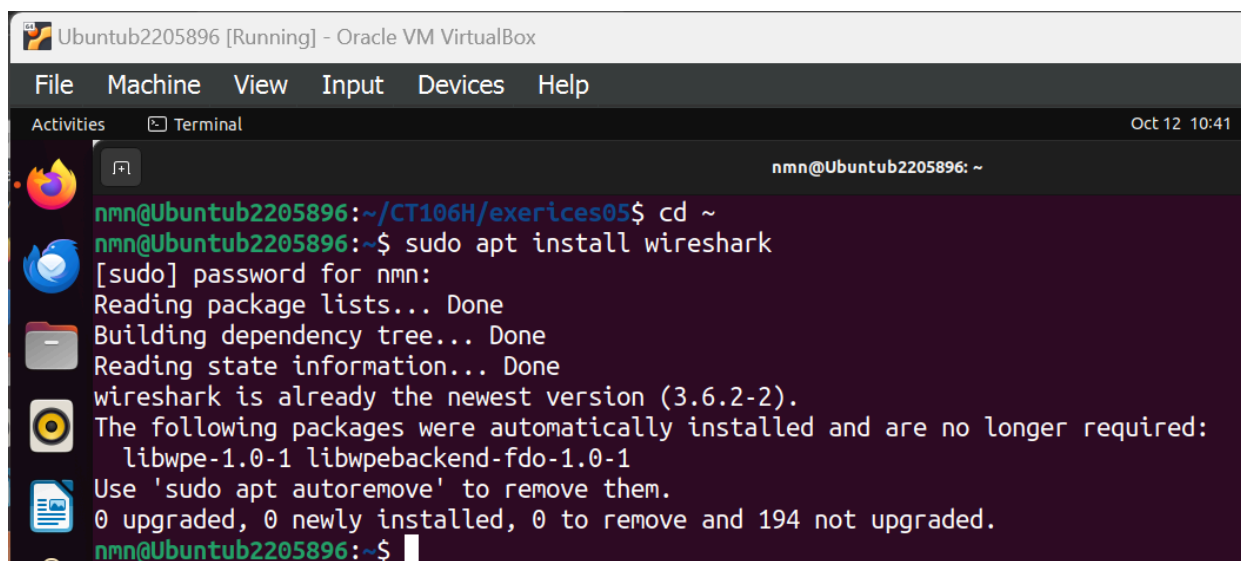
```

Use Wireshark:

Step 1 - install Wireshark:

Answer:

\$ sudo apt install wireshark



```
nmn@Ubuntub2205896:~/CT106H/exerices05$ cd ~
nmn@Ubuntub2205896:~$ sudo apt install wireshark
[sudo] password for nmn:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
wireshark is already the newest version (3.6.2-2).
The following packages were automatically installed and are no longer required:
  libwpe-1.0-1 libwpebackend-fdo-1.0-1
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 194 not upgraded.
nmn@Ubuntub2205896:~$
```

Step 2 - start the network: Reuse the network of exercise 5:

```
$ cd CT106H/exercise05
```

```
$ kathara lstart
```

Step 3 - run these commands:

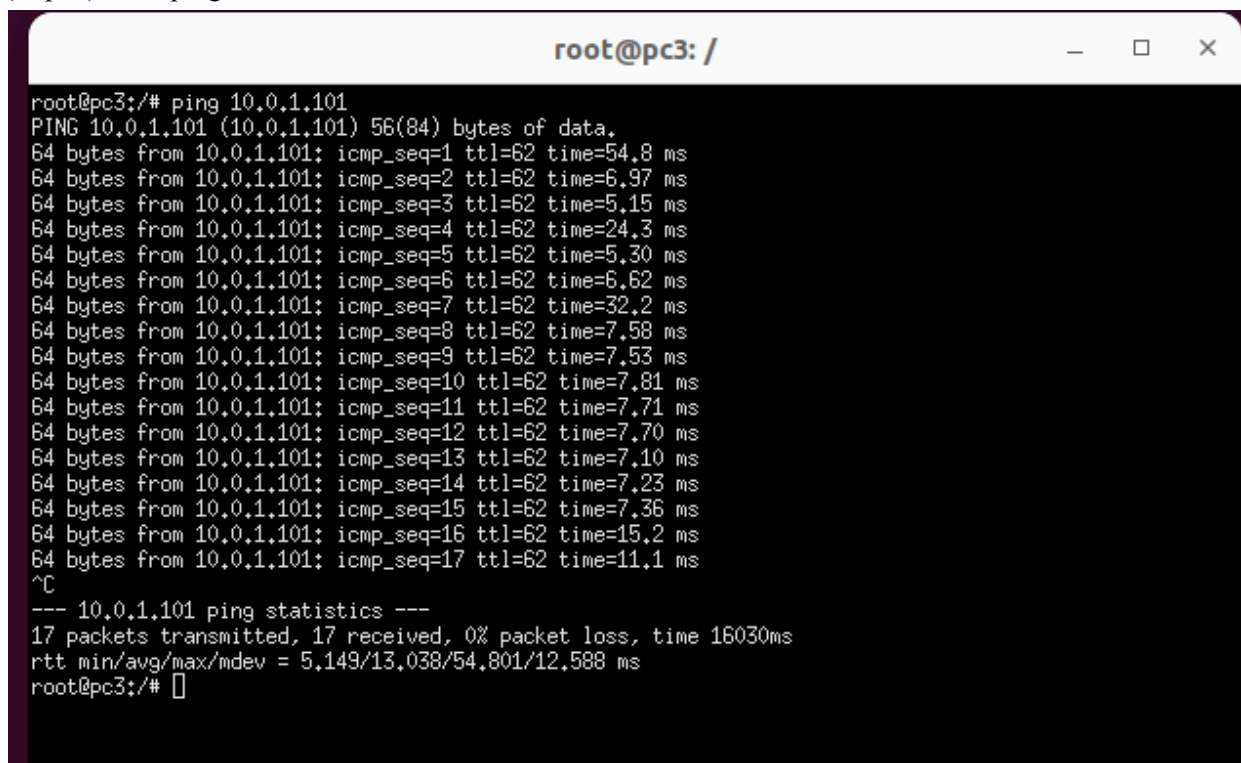
```
(In pc2) # tcpdump -s 1536 -w /hostlab/Ex5_pc2.pcap
```

```
(In router1) # tcpdump -s 1536 -w /hostlab/BT5_router1.pcap
```

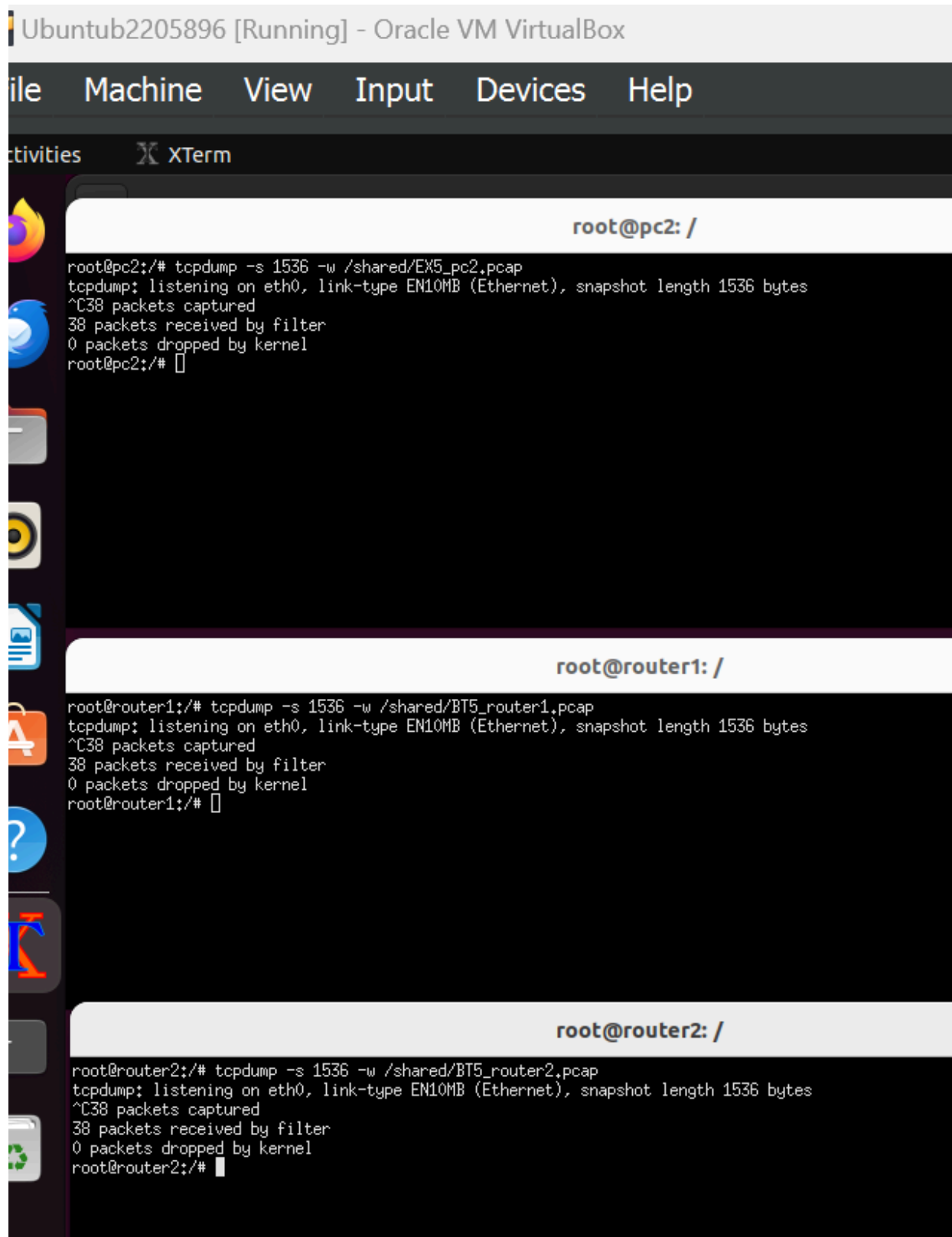
```
(In router2) # tcpdump -s 1536 -w /hostlab/BT5_router2.pcap
```

Run pc3 and stop these command by Ctrl + C in order: pc3, pc2, router1, router 2:

```
(In pc3) # ping 10.0.1.101
```



```
root@pc3: /
root@pc3:/# ping 10.0.1.101
PING 10.0.1.101 (10.0.1.101) 56(84) bytes of data.
64 bytes from 10.0.1.101: icmp_seq=1 ttl=62 time=54.8 ms
64 bytes from 10.0.1.101: icmp_seq=2 ttl=62 time=6.97 ms
64 bytes from 10.0.1.101: icmp_seq=3 ttl=62 time=5.15 ms
64 bytes from 10.0.1.101: icmp_seq=4 ttl=62 time=24.3 ms
64 bytes from 10.0.1.101: icmp_seq=5 ttl=62 time=5.30 ms
64 bytes from 10.0.1.101: icmp_seq=6 ttl=62 time=6.62 ms
64 bytes from 10.0.1.101: icmp_seq=7 ttl=62 time=32.2 ms
64 bytes from 10.0.1.101: icmp_seq=8 ttl=62 time=7.58 ms
64 bytes from 10.0.1.101: icmp_seq=9 ttl=62 time=7.53 ms
64 bytes from 10.0.1.101: icmp_seq=10 ttl=62 time=7.81 ms
64 bytes from 10.0.1.101: icmp_seq=11 ttl=62 time=7.71 ms
64 bytes from 10.0.1.101: icmp_seq=12 ttl=62 time=7.70 ms
64 bytes from 10.0.1.101: icmp_seq=13 ttl=62 time=7.10 ms
64 bytes from 10.0.1.101: icmp_seq=14 ttl=62 time=7.23 ms
64 bytes from 10.0.1.101: icmp_seq=15 ttl=62 time=7.36 ms
64 bytes from 10.0.1.101: icmp_seq=16 ttl=62 time=15.2 ms
64 bytes from 10.0.1.101: icmp_seq=17 ttl=62 time=11.1 ms
^C
--- 10.0.1.101 ping statistics ---
17 packets transmitted, 17 received, 0% packet loss, time 16030ms
rtt min/avg/max/mdev = 5.149/13.038/54.801/12.588 ms
root@pc3:/#
```



Ubuntub2205896 [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

Activities XTerm

root@pc2: /

```
root@pc2:/# tcpdump -s 1536 -w /shared/EX5_pc2.pcap
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 1536 bytes
^C38 packets captured
38 packets received by filter
0 packets dropped by kernel
root@pc2:/#
```

root@router1: /

```
root@router1:/# tcpdump -s 1536 -w /shared/BT5_router1.pcap
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 1536 bytes
^C38 packets captured
38 packets received by filter
0 packets dropped by kernel
root@router1:/#
```

root@router2: /

```
root@router2:/# tcpdump -s 1536 -w /shared/BT5_router2.pcap
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 1536 bytes
^C38 packets captured
38 packets received by filter
0 packets dropped by kernel
root@router2:/#
```

Step 4 - open exercise5_pc2.pcap using Wireshark, select the frame #3 and answer the following questions:

- Size of frame in bytes? Answer: 98 bytes

The screenshot shows the Wireshark interface with the file 'EX5_pc2.pcap' open. The packet list on the left shows 10 frames of ICMP Echo (ping) traffic. Frame 3 is selected, which is an ICMP Echo (ping) request from 10.0.2.101 to 10.0.1.101, 98 bytes in length. The packet details pane on the right shows the frame structure: Ethernet II (Src: 32:ff:32:e0:d9:b2, Dst: f6:ff:ba:d4:07:08), Internet Protocol Version 4 (Src: 10.0.2.101, Dst: 10.0.1.101), and Internet Control Message Protocol. The packet bytes pane at the bottom shows the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	10.0.2.101	10.0.1.101	ICMP	98	Echo (ping) request
2	0.000081	10.0.1.101	10.0.2.101	ICMP	98	Echo (ping) reply
3	0.979300	10.0.2.101	10.0.1.101	ICMP	98	Echo (ping) request
4	0.979333	10.0.1.101	10.0.2.101	ICMP	98	Echo (ping) reply
5	1.980810	10.0.2.101	10.0.1.101	ICMP	98	Echo (ping) request
6	1.980846	10.0.1.101	10.0.2.101	ICMP	98	Echo (ping) reply
7	2.981482	10.0.2.101	10.0.1.101	ICMP	98	Echo (ping) request
8	2.981646	10.0.1.101	10.0.2.101	ICMP	98	Echo (ping) reply
9	3.983622	10.0.2.101	10.0.1.101	ICMP	98	Echo (ping) request
10	3.983652	10.0.1.101	10.0.2.101	ICMP	98	Echo (ping) reply

Frame 3: 98 bytes on wire (784 bits), 98 bytes captured (784 bits)

- Ethernet II, Src: 32:ff:32:e0:d9:b2 (32:ff:32:e0:d9:b2), Dst: f6:ff:ba:d4:07:08 (f6:ff:ba:d4:07:08)
- Internet Protocol Version 4, Src: 10.0.2.101, Dst: 10.0.1.101
- Internet Control Message Protocol

```
0000  f6 ff ba d4 07 08 32 ff 32 e0 d9 b2 08 00 45 00  ....2. 2....E.
0010  00 54 a8 ee 40 00 3e 01 7b f1 0a 00 02 65 0a 00  .T. @.> {....e.
0020  01 65 08 00 05 3f 00 03 00 02 ff fe 09 67 00 00  .e...?.. ....g.
0030  00 00 23 83 07 00 00 00 00 00 10 11 12 13 14 15  ..#.....
0040  16 17 18 19 1a 1b 1c 1d 1e 1f 20 21 22 23 24 25  ..... !"#$$%
0050  26 27 28 29 2a 2b 2c 2d 2e 2f 30 31 32 33 34 35  &'()*+,-./012345
0060  36 37                                     67
```

- Select Header Internet Control Message Protocol → Which protocol is using? On which layer of the OSI model does this protocol operate? What is the content of the message? How long is this message in bytes?

EX5_pc2.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	10.0.2.101	10.0.1.101	ICMP	98	Echo (ping) request id=0x0003, ...
2	0.000081	10.0.1.101	10.0.2.101	ICMP	98	Echo (ping) reply id=0x0003, ...
3	0.979300	10.0.2.101	10.0.1.101	ICMP	98	Echo (ping) request id=0x0003, ...
4	0.979333	10.0.1.101	10.0.2.101	ICMP	98	Echo (ping) reply id=0x0003, ...
5	1.980810	10.0.2.101	10.0.1.101	ICMP	98	Echo (ping) request id=0x0003, ...
6	1.980846	10.0.1.101	10.0.2.101	ICMP	98	Echo (ping) reply id=0x0003, ...
7	2.981482	10.0.2.101	10.0.1.101	ICMP	98	Echo (ping) request id=0x0003, ...
8	2.981646	10.0.1.101	10.0.2.101	ICMP	98	Echo (ping) reply id=0x0003, ...
9	3.983622	10.0.2.101	10.0.1.101	ICMP	98	Echo (ping) request id=0x0003, ...
10	3.983652	10.0.1.101	10.0.2.101	ICMP	98	Echo (ping) reply id=0x0003, ...
11	4.987290	10.0.2.101	10.0.1.101	ICMP	98	Echo (ping) request id=0x0003, ...
12	4.987324	10.0.1.101	10.0.2.101	ICMP	98	Echo (ping) reply id=0x0003, ...
13	5.059152	f6:ff:ba:d4:07:08	32:ff:32:e0:d9:b2	ARP	42	Who has 10.0.1.1? Tell 10.0.1.1...

Type: 8 (Echo (ping) request)
 Code: 0
 Checksum: 0x053f [correct]
 [Checksum Status: Good]
 Identifier (BE): 3 (0x0003)
 Identifier (LE): 768 (0x0300)
 Sequence Number (BE): 2 (0x0002)
 Sequence Number (LE): 512 (0x0200)
 [Response frame: 4]
 Timestamp from icmp data: Oct 12, 2024 11:45:51.000000000 +07
 [Timestamp from icmp data (relative): 0.496225000 seconds]
 Data (48 bytes)
 Data: 2383070000000000101112131415161718191a1b1c1d1e1f202122232425262728292a2b...
 [Length: 48]

```

0000  f6 ff ba d4 07 08 32 ff 32 e0 d9 b2 08 00 45 00  .T..@.>..{...e..
0010  00 54 a8 ee 40 00 3e 01 7b f1 0a 00 02 65 0a 00  .e...?...g...
0020  01 65 08 00 05 3f 00 03 00 02 ff fe 09 67 00 00  .#.....
0030  00 00 23 83 07 00 00 00 00 00 10 11 12 13 14 15  ..... !"#$$%
0040  16 17 18 19 1a 1b 1c 1d 1e 1f 20 21 22 23 24 25  &'()*+,-./012345
0050  26 27 28 29 2a 2b 2c 2d 2e 2f 30 31 32 33 34 35  67
0060  36 37
  
```

Answer:

- Protocol: ICMP

- Layer: Network Layer

- Content of the message:

2383070000000000101112131415161718191a1b1c1d1e1f202122232425262728292a2b...

- Message Long: 48 bytes

- Select Header Internet Protocol Version 4 → What are the IP addresses of the source and destination hosts?

EX5_pc2.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	10.0.2.101	10.0.1.101	ICMP	98	Echo (ping) request id=0x0003, id=0x0003,
2	0.000081	10.0.1.101	10.0.2.101	ICMP	98	Echo (ping) reply id=0x0003,
3	0.979300	10.0.2.101	10.0.1.101	ICMP	98	Echo (ping) request id=0x0003,
4	0.979333	10.0.1.101	10.0.2.101	ICMP	98	Echo (ping) reply id=0x0003,
5	1.980810	10.0.2.101	10.0.1.101	ICMP	98	Echo (ping) request id=0x0003,
6	1.980846	10.0.1.101	10.0.2.101	ICMP	98	Echo (ping) reply id=0x0003,
7	2.981482	10.0.2.101	10.0.1.101	ICMP	98	Echo (ping) request id=0x0003,
8	2.981646	10.0.1.101	10.0.2.101	ICMP	98	Echo (ping) reply id=0x0003,
9	3.983622	10.0.2.101	10.0.1.101	ICMP	98	Echo (ping) request id=0x0003,
10	3.983652	10.0.1.101	10.0.2.101	ICMP	98	Echo (ping) reply id=0x0003,
11	4.987290	10.0.2.101	10.0.1.101	ICMP	98	Echo (ping) request id=0x0003,
12	4.987324	10.0.1.101	10.0.2.101	ICMP	98	Echo (ping) reply id=0x0003,
13	5.059152	f6:ff:ba:d4:07:08	32:ff:32:e0:d9:b2	ARP	42	Who has 10.0.1.1? Tell 10.0.1.1

Frame 3: 98 bytes on wire (784 bits), 98 bytes captured (784 bits)

Ethernet II, Src: 32:ff:32:e0:d9:b2 (32:ff:32:e0:d9:b2), Dst: f6:ff:ba:d4:07:08 (f6:ff:ba:d4:07:08)

Internet Protocol Version 4, Src: 10.0.2.101, Dst: 10.0.1.101

0100 = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 84
Identification: 0xa8ee (43246)
Flags: 0x40, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 62
Protocol: ICMP (1)
Header Checksum: 0x7bf1 [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.0.2.101

0000 f6 ff ba d4 07 08 32 ff 32 e0 d9 b2 08 00 45 002. 2.....E.
0010 00 54 a8 ee 40 00 3e 01 7b f1 0a 00 02 65 0a 00 .T..@.>. {....e..
0020 01 65 08 00 05 3f 00 03 00 02 ff fe 09 67 00 00 .e..?..g..
0030 00 00 23 83 07 00 00 00 00 00 10 11 12 13 14 15 ..#.....
0040 16 17 18 19 1a 1b 1c 1d 1e 1f 20 21 22 23 24 25!"#\$%
0050 26 27 28 29 2a 2b 2c 2d 2e 2f 30 31 32 33 34 35 &'()*+,-./012345
0060 36 37 67

Answer:

- Source Address: 10.0.2.101
- Destination Address: 10.0.1.101

- Select Header Internet Protocol Version 4 → What is the length of the IP packet header? What fields does the Header include? How long is each field (Bytes) ?

The screenshot shows the Wireshark interface with a packet capture of ICMP Echo (ping) requests and replies. The packet list shows several frames, and the packet details pane for Frame 3 is expanded, showing the Internet Protocol Version 4 header fields.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	10.0.2.101	10.0.1.101	ICMP	98	Echo (ping) request id=0x0003,
2	0.000081	10.0.1.101	10.0.2.101	ICMP	98	Echo (ping) reply id=0x0003,
3	0.979300	10.0.2.101	10.0.1.101	ICMP	98	Echo (ping) request id=0x0003,
4	0.979333	10.0.1.101	10.0.2.101	ICMP	98	Echo (ping) reply id=0x0003,
5	1.980810	10.0.2.101	10.0.1.101	ICMP	98	Echo (ping) request id=0x0003,
6	1.980846	10.0.1.101	10.0.2.101	ICMP	98	Echo (ping) reply id=0x0003,
7	2.981482	10.0.2.101	10.0.1.101	ICMP	98	Echo (ping) request id=0x0003,
8	2.981646	10.0.1.101	10.0.2.101	ICMP	98	Echo (ping) reply id=0x0003,
9	3.983622	10.0.2.101	10.0.1.101	ICMP	98	Echo (ping) request id=0x0003,
10	3.983652	10.0.1.101	10.0.2.101	ICMP	98	Echo (ping) reply id=0x0003,

Frame 3: 98 bytes on wire (784 bits), 98 bytes captured (784 bits)
 Ethernet II, Src: 32:ff:32:e0:d9:b2 (32:ff:32:e0:d9:b2), Dst: f6:ff:ba:d4:07:08 (f6:ff:ba:d4:07:08)
 Internet Protocol Version 4, Src: 10.0.2.101, Dst: 10.0.1.101

- 0100 = Version: 4
- 0101 = Header Length: 20 bytes (5)
- Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
- Total Length: 84
- Identification: 0xa8ee (43246)
- Flags: 0x40, Don't fragment
- ...0 0000 0000 0000 = Fragment Offset: 0
- Time to Live: 62
- Protocol: ICMP (1)
- Header Checksum: 0x7bf1 [validation disabled]
 [Header checksum status: Unverified]
- Source Address: 10.0.2.101
- Destination Address: 10.0.1.101

Internet Control Message Protocol

Answer:

- IP packet header length: 20 bytes
- Header include fields:
 - + Version: 4 bits
 - + Header Length: 4 bits
 - +Type of Service: 1 byte
 - +Total Length: 2 bytes
 - +Identification: 2 bytes
 - +Flags + Fragment Offset: 2 bytes
 - +Time to Live: 1 byte
 - +Protocol: 1 byte
 - +Header Checksum: 2 bytes
 - +Source Address: 4 bytes
 - +Destination Address: 4 bytes

- Select Header Internet Protocol Version 4 → What is the length of the Total Length field (Bytes)?

The screenshot shows the Wireshark interface with the packet list and details pane. The packet list shows a series of ICMP Echo (ping) requests and replies between 10.0.2.101 and 10.0.1.101. The details pane for packet 3 shows the Internet Protocol Version 4 header with a Total Length of 84 bytes.

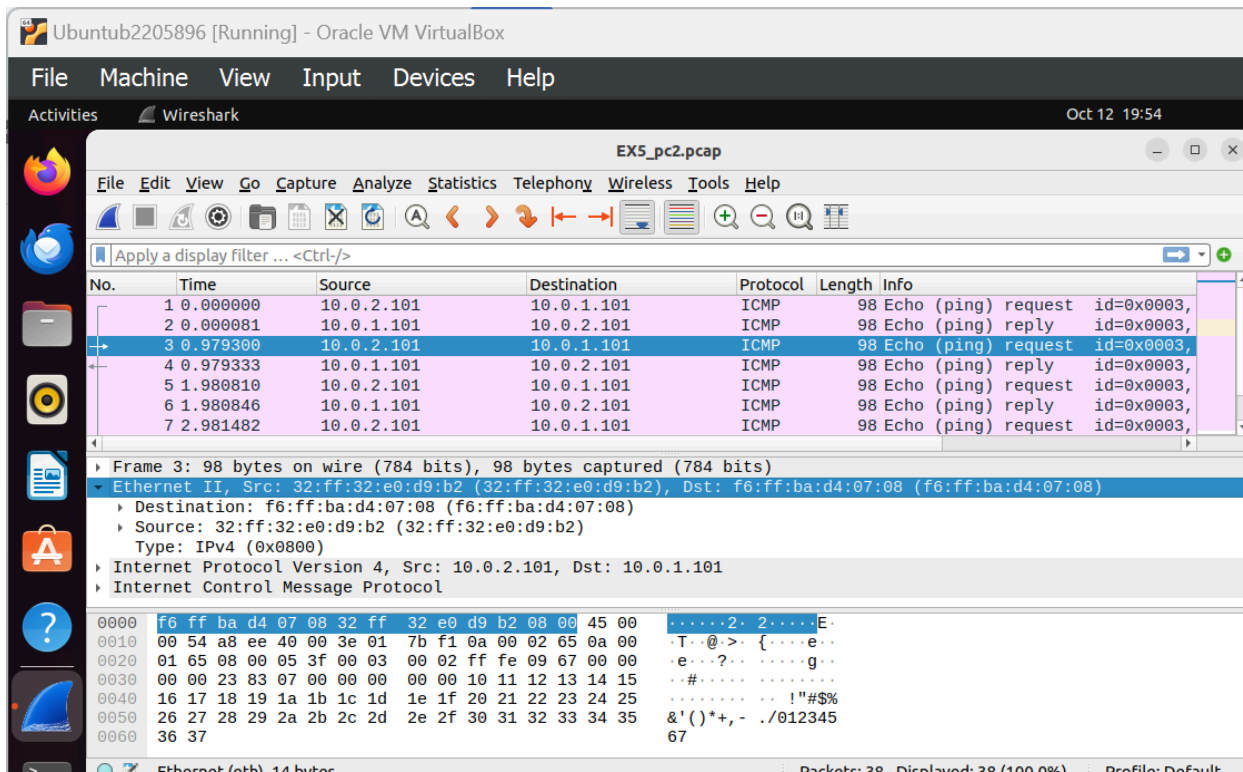
No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	10.0.2.101	10.0.1.101	ICMP	98	Echo (ping) request id=0x0003, id=0x0003
2	0.000081	10.0.1.101	10.0.2.101	ICMP	98	Echo (ping) reply id=0x0003, id=0x0003
3	0.979300	10.0.2.101	10.0.1.101	ICMP	98	Echo (ping) request id=0x0003, id=0x0003
4	0.979333	10.0.1.101	10.0.2.101	ICMP	98	Echo (ping) reply id=0x0003, id=0x0003
5	1.980810	10.0.2.101	10.0.1.101	ICMP	98	Echo (ping) request id=0x0003, id=0x0003
6	1.980846	10.0.1.101	10.0.2.101	ICMP	98	Echo (ping) reply id=0x0003, id=0x0003
7	2.981482	10.0.2.101	10.0.1.101	ICMP	98	Echo (ping) request id=0x0003, id=0x0003
8	2.981646	10.0.1.101	10.0.2.101	ICMP	98	Echo (ping) reply id=0x0003, id=0x0003
9	3.983622	10.0.2.101	10.0.1.101	ICMP	98	Echo (ping) request id=0x0003, id=0x0003
10	3.983652	10.0.1.101	10.0.2.101	ICMP	98	Echo (ping) reply id=0x0003, id=0x0003
11	4.007200	10.0.2.101	10.0.1.101	ICMP	98	Echo (ping) request id=0x0003, id=0x0003

Frame 3: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface 0
Ethernet II, Src: 32:ff:32:e0:d9:b2 (32:ff:32:e0:d9:b2), Dst: f6:ff:ba:d4:07:08 (f6:ff:ba:d4:07:08)
Internet Protocol Version 4, Src: 10.0.2.101, Dst: 10.0.1.101
0100 = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 84
Identification: 0xa8ee (43246)
Flags: 0x40, Don't fragment
... 0000 0000 0000 = Fragment Offset: 0
Time to Live: 62
Protocol: ICMP (1)
Header Checksum: 0x7bf1 [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.0.2.101
Destination Address: 10.0.1.101
Internet Control Message Protocol

0010 00 54 a8 ee 40 00 3e 01 7b f1 0a 00 02 65 0a 00 ..@.> {...e..
0020 01 65 08 00 05 3f 00 03 00 02 ff fe 09 67 00 00 .e...?.....g..
0030 00 00 23 83 07 00 00 00 00 00 10 11 12 13 14 15 .#.....
0040 16 17 18 19 1a 1b 1c 1d 1e 1f 20 21 22 23 24 25 !"#\$\$%
0050 26 27 28 29 2a 2b 2c 2d 2e 2f 30 31 32 33 34 35 &'()*+,-./012345
0060 36 37 67

Answer: 84 bytes

- Select Header Ethernet II → What are the MAC addresses of the source and the destination hosts?



Answer:

- Destination: f6:ff:ba:d4:07:08 (f6:ff:ba:d4:07:08)
- Source: 32:ff:32:e0:d9:b2 (32:ff:32:e0:d9:b2)

- Select Header Ethernet II → What is the Type value?

Answer:

- IPv4 (0x0800)

END